



Bioelectricity of non-excitable cells and multicellular pattern memories: Biophysical modeling

Javier Cervera^a, Michael Levin^{b,c}, Salvador Mafe^{a,*}

^a Dept. Termodinàmica, Facultat de Física, Universitat de València, E-46100 Burjassot, Spain

^b Department of Biology and Allen Discovery Center at Tufts University, Medford, USA

^c Wyss Institute for Biologically Inspired Engineering, Harvard University, Boston, USA

ARTICLE INFO

Article history:

Received 21 April 2022

Received in revised form 17 October 2022

Accepted 9 December 2022

Available online 22 December 2022

Editor: Massimo Vergassola

Keywords:

Bioelectricity

Electric potential

Transcription

Patterns

Biophysical modeling

ABSTRACT

Cells must coordinate their individual activities toward multicellular goals, which requires efficient information processing mechanisms. Bioelectrical signals encode instructive rules at multiple scales, from the individual cell to the tissue and organ levels, because patterns of cell potentials are locally coupled to transcription and morphogenesis via biochemical downstream processes. We review here biophysical models that suggest how bi-stable and oscillatory bioelectrical memories defined at the single-cell and multicellular scales can constitute pattern memories that are instructive for morphological outcomes. Multicellular aggregates are open to the external microenvironment and typically show spatio-temporal distributions of potassium and calcium ions, neurotransmitters, and specific transcription activators that are correlated with electric potential patterns. This correlation results in patterns composed of dynamic subsystems (modules) with cells that share the same bioelectrical state. The particular integration–segregation topology of the different modules defines a multicellular pattern memory. By acting on these separate modules and their particular integration, pattern memories can be retrieved or externally rewritten, with morphological consequences. Thus, the simulations give further support to recent experimental findings and suggest new opportunities for external actions at the intermediate scale characteristic of endogenous multicellular fields.

© 2022 Elsevier B.V. All rights reserved.

Contents

1. Introduction.....	2
2. Bioelectrical model.....	4
3. Bi-stable single-cell memory.....	6
4. Coupling of bioelectricity with transcription.....	7
5. Oscillatory single-cell memory.....	8
6. Multicellular pattern memories.....	11
6.1. The community effect.....	11
6.2. Intercellular gap junction plasticity.....	13
6.3. Bioelectrical pattern memories: examples.....	15
6.4. Bi-stable and oscillatory pattern memories.....	18
6.5. Oscillatory phenomena and synchronization.....	20

* Corresponding author.

E-mail address: smafe@uv.es (S. Mafe).

6.6. Extensions of bioelectrical model.....	22
7. Discussion, open problems, and perspectives.....	25
CRedit authorship contribution statement.....	28
Declaration of competing interest.....	28
Acknowledgments.....	28
References.....	28

1. Introduction

Cells are continuously responding to environmental signals. The cell membrane acts as an information processing interface that incorporates a multitude of sensing and actuating channel proteins [1,2]. A similar claim has been made about the tissue-level membrane, the epithelium [3]. The chemical, mechanical, and electrical signals from the microenvironment, together with the intercellular communication, allow cells to coordinate their individual states in collective functional states that do not exist at the single-cell level. While a systems-level view of signal integration is central to pattern dynamics and developmental approaches [4–6], only biochemical and biomechanical signaling processes are usually emphasized. However, *multicellular bioelectrical patterns* can be instructive in development, regeneration, and tumorigenesis [7–15], as shown in Fig. 1a–c. Indeed, tissues form bioelectrical networks reminiscent of neural networks where the spatio-temporal membrane potentials distributions are regulated by the individual cell states and the intercellular connectivity provided by the intercellular gap junctions (Fig. 1c). The resulting bioelectric gradients modulate the transport of the neurotransmitters and signaling ions and molecules that influence gene expression and morphogenesis (Fig. 1d).

Here, we use the term *pattern memory* for designating those instructive bioelectric states of non-neural multicellular aggregates that are stable for developmentally relevant times and can be retrieved by appropriate external actions [9,12]. These patterns can also be envisaged as regions of morphospace in model biosystems [8,15]. The transitions between these regions can be induced by acting on individual cell potentials as well as on the relative potentials between adjacent cells [16].

Experimentally, the spatial distributions of calcium ions, neurotransmitters, and other transcription activators (and ultimately, morphogenetic changes) are correlated with cell bioelectricity because of a range of transduction mechanisms: the electrical potential regulates morphogen electrophoresis through intercellular gap junctions, voltage-sensitive phosphatases and ion transporters, cytoskeletal changes, KRAS clustering, and voltage-dependent neurotransmitters release [8]. In this way, the cell potential modulates transcription locally via biochemical and mechanical downstream processes and thus different phenotypes can emerge from controlled alterations in bioelectrical regulators such as ion channels and intercellular junctions [8–10,12–15].

Modularity and integration among modules are central to development and emerge at all levels of biological organization; in the case of bioelectric signals, they have been shown to encode individual cell states (stem cell differentiation), pre-patterns of tissue distribution (craniofacial patterning), organ identity (eye induction), or a generic “generate whatever organ normally belongs at this location” signal [9,10,17–19]. Understanding how these different meanings are assigned to bioelectric states in various context is a major open question for evolutionary developmental biology and biomedicine [9,16,18,20,21].

How can different multicellular modules encode bioelectrical pattern memories instructive for development and regeneration? How can cell-level bioelectric patterns coexist with tissue/organ-level ones? We review here, offering new perspectives and suggesting practical applications, a minimal biophysical model that leads to *bi-stable* and *oscillatory* memories both at the *single-cell* and *multicellular* levels. The spatio-temporal patterns that encode morphological outcomes can be obtained by allowing the cells to dynamically transit between different bioelectrical states acting as memories. Because the individual cell potentials are correlated with the multicellular field, different *dynamic modules* defined by subsystems of cells with similar bioelectrical state can coexist in a large multicellular aggregate [22–24]. The particular *integration-segregation topology* of the different modules defines a *multicellular pattern memory*. Importantly, these pattern memories can allow an operational control based on mean-field potentials, as an alternative to addressing the molecular characteristics of individual cells — a possibility that offers a much-needed explanation of the ability of bioelectric states to encode coarse-grained features, for example entire organs such as the eye or limb [20]. As suggested by Fig. 1, experimental examples of our systems-level approach are bioelectrically-regulated two-head planarian worms [9,12], the transplant of ion channel-expressing tissues in morphogenetic control mechanisms [13,15], and the local cell depolarization as an early step in the tumorigenesis of tadpoles injected with human oncogenes [14,22].

The biophysical models described are based on the cell potential V , defined as the electrical potential difference between the cell inside and the external microenvironment. High and low absolute values of V correspond to the polarized (*pol*) and depolarized (*dep*) *individual cell states* [7,8,12,25–29], which are regulated by the *dynamic balance* between channel families whose conductances aim at polarizing or depolarizing the cell, respectively. At the multicellular level, the intercellular junction conductances modulate the *system connectivity* [7,8,30–32].

Note that the absolute value of V is decreased over a significant part of the cell cycle [14,16,26] and the differences observed among the distinct cells of Fig. 2 suggest an experimental relationship between bioelectricity and transcription

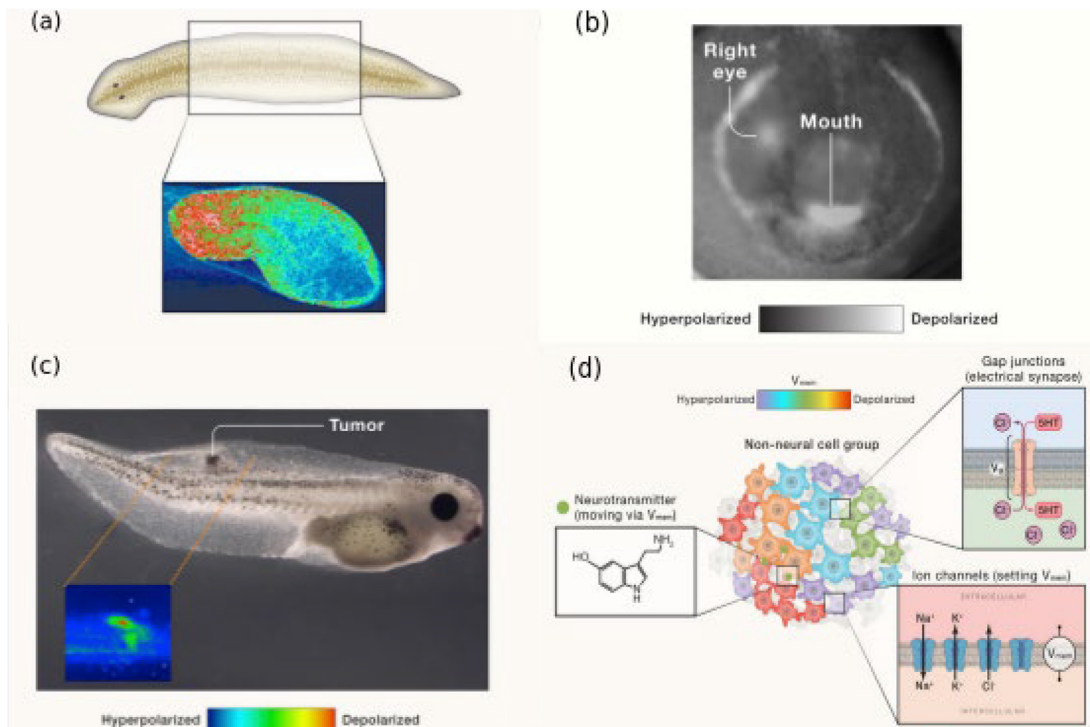


Fig. 1. System-level bioelectric patterns showing that *depolarized* (low absolute values of electric potential V) cells coexist with neighboring regions of *polarized* (high absolute values of V) cells in development, regeneration, and tumorigenesis processes: (a) tissue cell potentials visualized by voltage-sensitive dyes showing subsystems of cells with similar polarization state (*modules*) that define the axial polarity of a planarian flatworm fragment. This bioelectrical pattern determines developmental and regenerative processes [12]. (b) Endogenous pre-patterns with clearly discernible electric modules are instructive for the face morphogenesis of early frog ectoderm because they are correlated with particular gene-expression regions (mouth, eyes, etc.). External alterations of the channel proteins expression provoke changes in the electrical pattern that eventually impact on the head anatomy [13]. (c) A spatial module of cells in early tumorigenesis may depolarize and electrically decouple from the neighboring multicellular aggregate, thus deviating from the general organogenesis plan, as shown in tadpoles locally injected with human oncogenes [14]. (d) Experimentally, cells are integrated into differently polarized regions that act as dynamic *bioelectric modules* whose particular topology shapes the instructive *pattern memory*. We aim here at describing theoretically these functional patterns.

Source: Reprinted figure from Ref. [9].

(Fig. 3). In particular, the different average V values obtained experimentally in normal and cancer cell populations could be related to the probabilities of finding these cells in the distinct cell cycle phases. While acting on V should have instructive consequences [33], the relationship between bioelectricity and metastatic processes is complex and it may be possible that not all cancer cells have depolarized cell potentials. Heterogeneity of cell potentials, caused by differential expression of ion channels, has been suggested for metastatic breast cancer cells [21]. It is possible that a mixture of depolarized and hyperpolarized cells coexisted in this case. In the above context, biophysical models of the coupling between bioelectricity with transcription are much needed.

Because of the intercellular connections, bioelectricity is crucial not only at the single-cell but also at the multicellular level. Indeed, the genetic code specifies proteins via gene-regulatory networks and, coupled with it, the bioelectric code can specify instructions for morphological outcomes via multicellular patterns of cell potentials (Fig. 3). This coupling provides pattern memories that offer opportunities for biomedical interventions based on large-scale units such as organs [8,33].

Pathways converting bioelectrical states into individual cell properties have been characterized extensively [33]. However, the representation of large-scale anatomy by tissue-level bioelectric distributions – the bioelectric code – is still poorly understood. Thus, here we focus on the bioelectrical patterns rather than on the associated downstream biochemical processes. For instance, the local calcium concentration, modulated by cell potentials and voltage-gated ion channels, is central to signaling waves in pancreatic islets [34] and regulates crucial processes such as proliferation, apoptosis, the cell cycle, wound healing, transcription, and synchronized oscillations [35–38]. We believe that this electrically-focused multicellular approach is timely and may offer new perspectives: while it is experimentally established that bioelectric signals can influence development, regeneration, and cancer, the coupling between the single-cell and multicellular levels is still under discussion and can benefit from simple models that emphasize the basic qualitative concepts. This is an important step toward future strategies for biomedical interventions that exploit modular bioelectric control.

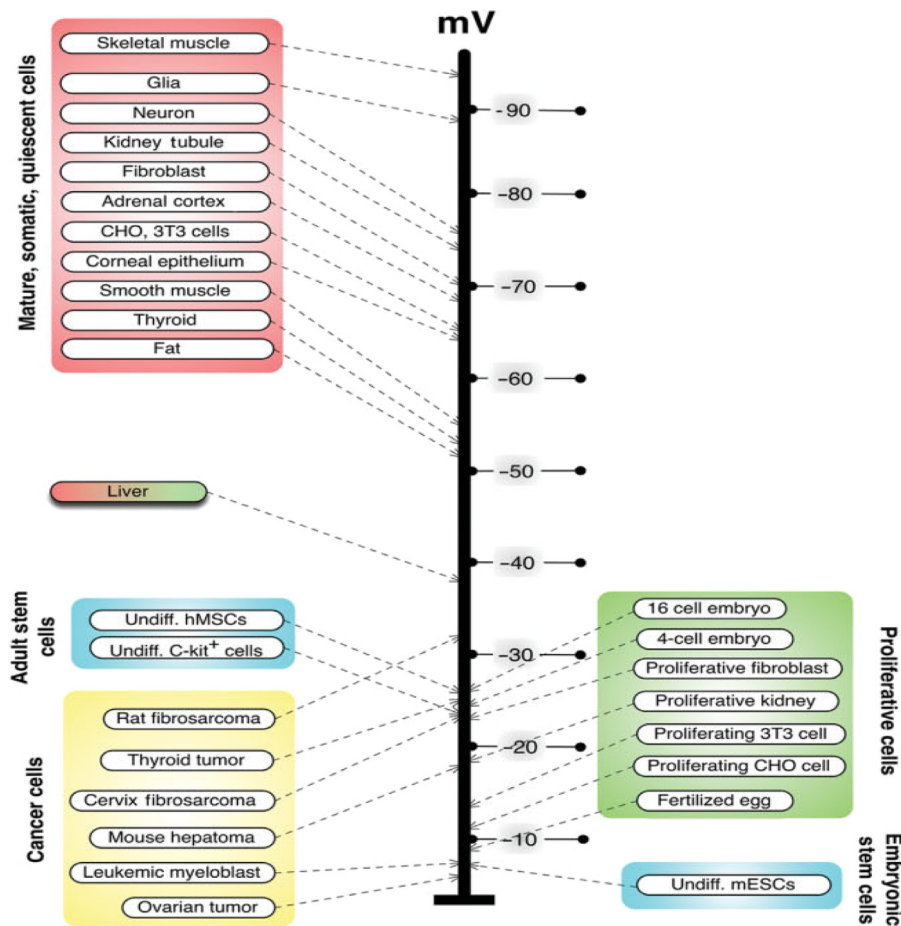


Fig. 2. Scheme of the membrane potential of different cells. Differentiated, quiescent cells tend to be typically hyperpolarized with respect to stem, tumor, and embryonic cells which are depolarized. For a single-cell, V is modulated by the conductance of the membrane ion channels so that acting on these channels should influence cell proliferation and plasticity.

Source: Reprinted figure from Ref. [33].

As explained above, this report has clear experimental motivations. However, we will emphasize here the biophysical concepts, putting every theoretical result in the relevant experimental context where it provides useful qualitative insights. Also, a sufficient number of references to the original literature will be given to motivate the interest of both fundamental and applied physicists.

2. Bioelectrical model

The model is based on two well-established experimental facts (Figs. 1–3):

- (i) regions of relatively depolarized cells can coexist with neighboring regions of polarized cells in the bioelectrical patterns characteristic of development, regeneration, and tumorigenesis [7,13,14,22]; the local potentials control crucial cell membrane functions and modulate the spatial distributions of signaling ions and molecules, thus influencing protein transcription locally [8,9]; and
- (ii) the dynamics of these multicellular patterns can be described in terms of average electric potentials. These potentials and the associated currents have morphological outcomes, as shown in the early studies by Jaffe and others on the role of ion transporters segregation and ionic currents in the establishment of developmental patterns [39]; see also more recent work on the primary axis of planaria worms and craniofacial and brain development in *Xenopus* embryos [7,9,12,15].

While most mathematical models for pattern formation and transcriptional control emphasize morphogen gradients and reaction–diffusion equations [40–42], bioelectrical and transcriptional networks are also coupled, as suggested by the experimental cell potentials of Fig. 2. Thus, the transcription of ion channel and gap junction proteins can in turn

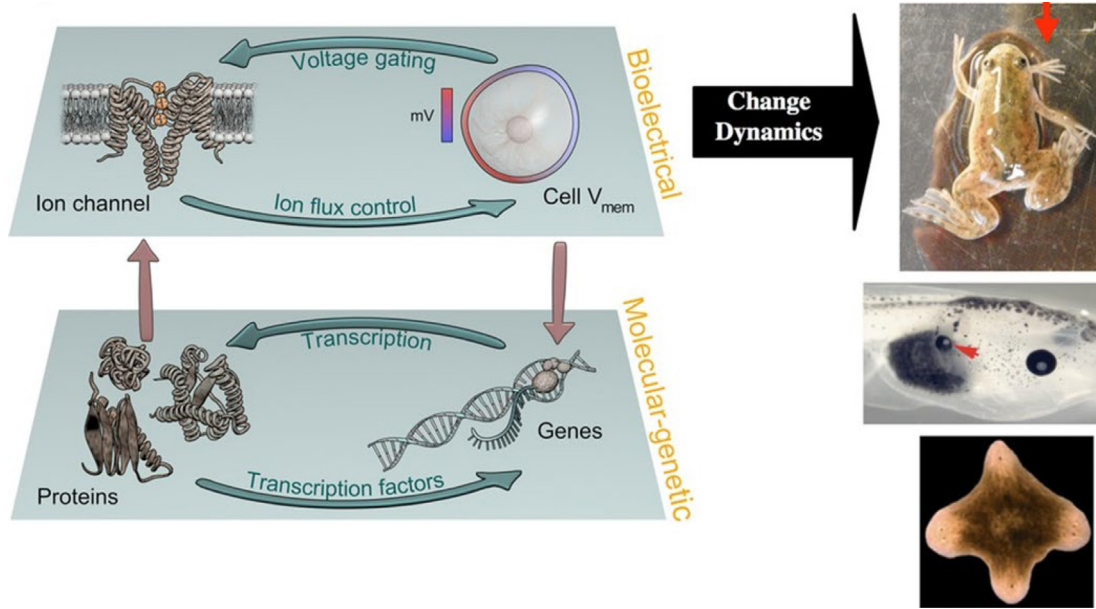


Fig. 3. The membrane potential that characterizes the cell bioelectrical state can establish instructive pre-patterns at the multicellular level. Although this bioelectrical magnitude is not a transcription factor itself, it influences transcription via second-messengers such as calcium and other signaling molecules that regulate downstream processes. Thus, many ion channels that are voltage-sensitive show feedback loops that couple together the bioelectrical and transcriptional layers (left). This coupling is instructive for large-scale patterning in vertebrate and invertebrate models (right), for instance ectopic limbs and eyes (red arrows), as well as heads at ectopic locations of flatworms. These experimental facts suggest new opportunities for biological reprogramming that do not involve genome editing. Left side diagrams by Alexis Pietak.

Source: Reprinted figure from Ref. [8].

be influenced by their own electrical activity [7,8,10,23], as schematically shown in Fig. 3. Also, electrical depolarization signals can propagate through gap junction networks over long distances and activate downstream signaling cascades during healing [43]. Here, we will focus on the electrical mechanisms themselves rather than on the comparatively slow downstream diffusional and transcriptional processes usually associated with bioelectricity [22,44].

Fig. 4 schematically shows a multicellular aggregate of potentials $V_i < 0$, $i = 1, 2, \dots, N$, that characterize the individual cell states together with the resulting electric potential pattern, which can be visualized by means of voltage-sensitive dyes [12,45,46]. The cell potentials are regulated by: (i) the *dynamic balance* between the currents through two generic *dep* and *pol* channels of each individual cell and (ii) the intercellular currents through the junctions between neighboring cells [32]. Similarly, only two types of counteracting voltage-gated channels regulate the neural cell excitability and the synaptic junctions modulate the plasticity of neural networks [1].

The channel proteins connect the cell inside and the microenvironment, thus regulating the *individual cell state*. While a multitude of membrane channels and pumps act in concert to establish the cell polarization state, we simulate the conductive effects of these ionic transporters by grouping them into two generic families of voltage-gated channels whose effective *dep* (G_{dep}) and *pol* (G_{pol}) conductances attempt to establish the respective equilibrium potentials E_{dep} and E_{pol} that characterize the *depolarized* and *polarized* cell states. These potentials depend on the external microenvironment and cell inside ionic concentrations, which can be assumed to be approximately constant because of the action of membrane ion pumps and transporters [1]. The model also includes the cell capacitance C as a membrane parameter that may influence the individual cell sensitivity to bioelectrical signals [10]. Typical values for the system parameters of the model equations (Fig. 4) are $G_k^0 = 10 - 1000$ pS ($k = pol, dep$), $E_{pol} = -60$ mV, $E_{dep} = 0$ mV, $C = 10 - 100$ pF, $z = 3$, and $V_{th} = -V_T = -26$ mV [1,8].

Note that in order to emphasize the local correlation between cell potentials and signaling ions and molecules, Fig. 4 schematically shows the concentration profile S of a hypothetical signaling ion (e.g. calcium) or molecule (e.g. serotonin) that is regulated by the electric potential difference between the head and tail. This feedback between biochemical transcription factors and bioelectrical cell potentials signals can occur in multiple ways, e.g. via the increased microenvironmental concentration of the signaling ion or molecule in the system polarized region or via the opening of a specific voltage-gated transporter in this region. Particular examples here are the local concentration increases due to the Donnan equilibrium phenomena caused by the cell membrane fixed charges [1] or the opening of calcium channels following cell depolarization [34], respectively. In addition to the cytosol, calcium is stored in the mitochondria and the endoplasmic reticulum, being released and transported through voltage-gated channels and intercellular junctions [35]. Also, the bioelectrical axial polarity regulates transcriptional factors that influence the downstream regenerative program determining the number and location of heads in planaria [12,46].

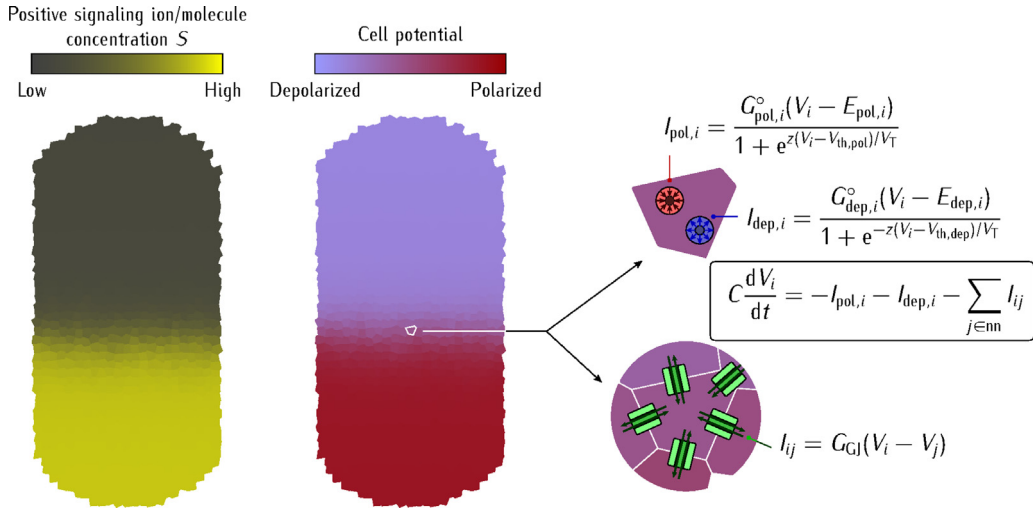


Fig. 4. Scheme of the multicellular model. The cell states are described by the potential V_i that changes with time t because of the single-cell currents $I_{k,i}$ through the conductances $G_{dep,i}$ and $G_{pol,i}$ of maximum values $G_{k,i}^0$ ($k = dep, pol$) and the intercellular current $I_{ij} = G_{GJ} \cdot (V_i - V_j)$ through the junction conductance G_{GJ} between the neighboring cells i and j [47]. $V_i - V_j$ is the intercellular potential difference, z is the channel gating charge, V_{th} is the threshold potential, and $V_T = RT/F$ is the thermal potential expressed in terms of the gas constant R , the temperature T , and the Faraday constant F [1]. The scheme also shows the concentration profile S of a positively charged ion or molecule that is regulated locally by the electric potential.

Source: Reprinted figure from Ref. [47].

3. Bi-stable single-cell memory

The biophysical model of Fig. 4 can give bi-stable cell states, as shown by the channels current–voltage curves (Fig. 5a) and cell potential V (Fig. 5b). The two stable values of the *membrane potential* $V(I = 0)$ that characterize the cell memory depend on the ratio of the channel conductances and the equilibrium potentials E_{dep} and E_{pol} that may change with the ionic concentrations of the inside and outside cell [1,22]. This difference between the polarized and depolarized cell states is a consequence of the minimal model of Fig. 4 and should not be so marked and abrupt in real systems because of the multitude of ion transporters that contribute to the cell membrane. However, it is an experimental fact that significantly different polarized and depolarized cell potentials, together with instructive potential windows, can be identified in model animals (Fig. 1); see also Refs. [7,12,33,48,49] for other clearly resolved cell polarization states. In addition, genuine bi-stable and two-level cell potentials have been found for different complements of ion channels in human cardiomyocytes [50], Purkinje neurons in vitro [51], model membranes at low extracellular potassium concentration [52], pancreatic islets [34], K^+ -mediated neurovascular communication [53], slow-wave oscillations in neurophysiology [54], and biosynthetic tissues [55,56]. In each case, a particular complement of ion channel proteins determines the bioelectrical state as a single-cell memory, a general characteristic incorporated in the bioelectrical model. Note also that biochemically-focused models that define the phenotype of a single cell in terms of its gene expression pattern may also describe each gene with a binary “on” or “off” state. In general, the implementation of memory in proteins can be based on positive and negative feedback mechanisms leading to bi-stability [57].

Having established experimentally observable cell depolarized and polarized states (Figs. 1, 4, and 5), we consider now how the cell can transit between the corresponding potentials. Fig. 6a shows that a transient weak perturbation due to an externally-induced change in the conductance ratio G_{pol}^0/G_{dep}^0 cannot change the cell polarization state after this stimulus is ceased. On the contrary, a strong enough perturbation can change the initial cell state (Fig. 6b). Experimentally, the *increase* in the *pol* channel conductance that gives the cell potential transition can be realized by the local microinjection of the specific mRNA [15] that encodes the *pol* channel protein. It is also possible to *decrease* the *dep* channel conductance by specific blockers [1,22]. Note that these two actions can increase the conductance ratio G_{pol}^0/G_{dep}^0 in Fig. 6. Alternatively, other cell potential transitions can be achieved by appropriate changes in the microenvironmental ionic concentrations that regulate the potential E_{pol} [1], as clearly shown in Fig. 5b. These external actions based on the pharmacological manipulation of the channels conductances and the ionic gradients across the cell membrane constitute experimentally plausible methods to modify the *bioelectrical memory state* encoded in cell potentials. Additional biophysical examples will be given in the next sections.

In summary, Figs. 5 and 6 suggest experimental procedures to implement bioelectrical memories based on the cell potential. Indeed, experimental modifications of the channel conductance ratio G_{pol}^0/G_{dep}^0 can be induced by mRNA microinjection and the introduction of specific channel blockers while the particular values of E_{pol} can also be controlled by changes in the salt concentration of the outside solution. Following these external actions, the cell potential can be

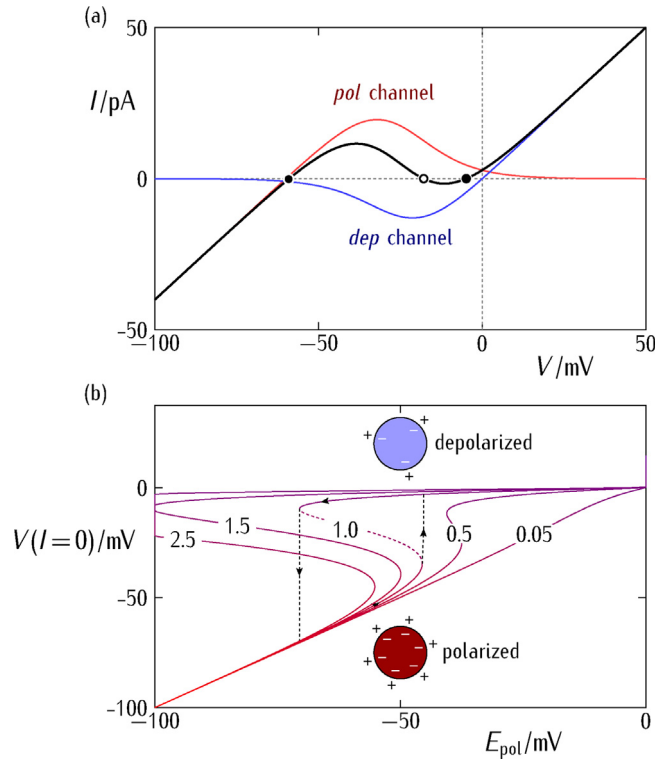


Fig. 5. Bi-stable cell potentials. (a) The *dep* and *pol* channel currents I_{dep} and I_{pol} of Fig. 4 together with the total current $I_{\text{dep}} + I_{\text{pol}}$ as functions of the potential V for the maximum effective conductances $G_{\text{pol}}^0 = G_{\text{dep}}^0 = 1000$ pS, the equilibrium potentials $E_{\text{pol}} = -60$ mV and $E_{\text{dep}} = 0$ mV, and the channel parameters $z = 3$ and $V_{\text{th}} = -V_T = -26$ mV. The two stable (black circles) and the unstable (white circle) membrane potentials correspond to the zero total current condition, $V(I = 0)$. (b) The membrane potential $V(I = 0)$ obtained from this condition as a function of the equilibrium potential E_{pol} parametrically in the conductance ratio $G_{\text{dep}}^0/G_{\text{pol}}^0$. The curve corresponding to $G_{\text{dep}}^0/G_{\text{pol}}^0 = 1$ shows the transitions between the depolarized and polarized stable potentials [22,28].

Source: Reprinted figure from Ref. [22].

switched between two *pol* and *dep* states that are stable after the external perturbation is removed, as shown in Figs. 5b and 6. Thus, the transient manipulation of target ion channels can lead to re-settings of cell bioelectrical states. Other bi-stability mechanisms based on gene activation under the control of morphogen levels above threshold values have also been proposed [58] but we will concentrate here on the correlation between transcriptional and bioelectrical networks.

4. Coupling of bioelectricity with transcription

We next extend the model of Fig. 4 to include the *feedback* between bioelectrical and transcriptional signals [59] schematically shown in Fig. 3. To this end, Fig. 7a considers the transcription of the channels that regulate the *pol* and *dep* cell states in terms of the mRNA and protein concentrations m_{pol} and m_{dep} and p_{pol} and p_{dep} , respectively. The rate constants for mRNA transcription and protein translation are $r_{m,k}$ and $r_{p,k}$ while their respective degradations rates are $d_{m,k}$ and $d_{p,k}$ ($k = \text{dep, pol}$). In this extended model relating bioelectricity to transcription, the voltage-gated channel conductance G_k depends not only on the cell potential V (Fig. 4) but also on the protein concentration p_k . This additional dependence can be incorporated using a Hill kinetics where the maximum channel conductance G_k^0 is now replaced by $G_k^* = G_k^0 p_k / (p_{0,k} + p_k)$ and the reference concentration $p_{0,k}$ corresponds to the half-maximum conductance $G_k^0/2$ [25]. The kinetic saturation of the *pol* and *dep* channel conductances can arise not only because of the finite transcription and translation rates of Fig. 7b but also because of other limitations due to protein trafficking to and insertion in the cell membrane. These additional effects, together with the possible channel blocking with specific ions and molecules, could also be accounted for by decreasing the maximum conductance G_k^0 [22].

The extended model of Fig. 7 includes the coupling between transcription and bioelectricity in terms of the voltage-dependent concentration S_k of a signaling ion or molecule, which results in effective mRNA transcription rates $r_{m,k}(V)$ that depend on the cell potential V , where $r_{m,k}(|V| = 0) = r_{m,k}^0/2$ ($k = \text{dep, pol}$). Because of the resulting feedback, V can regulate the protein concentrations p_k and thus the channel conductances G_k through *transcription*. In turn, the conductances G_k can regulate V through *bioelectricity*, as shown in Figs. 3, 6, and 7. It is important to realize that the model of Fig. 7 explicitly incorporates this double transcriptional/post-translational control of channel protein expression.

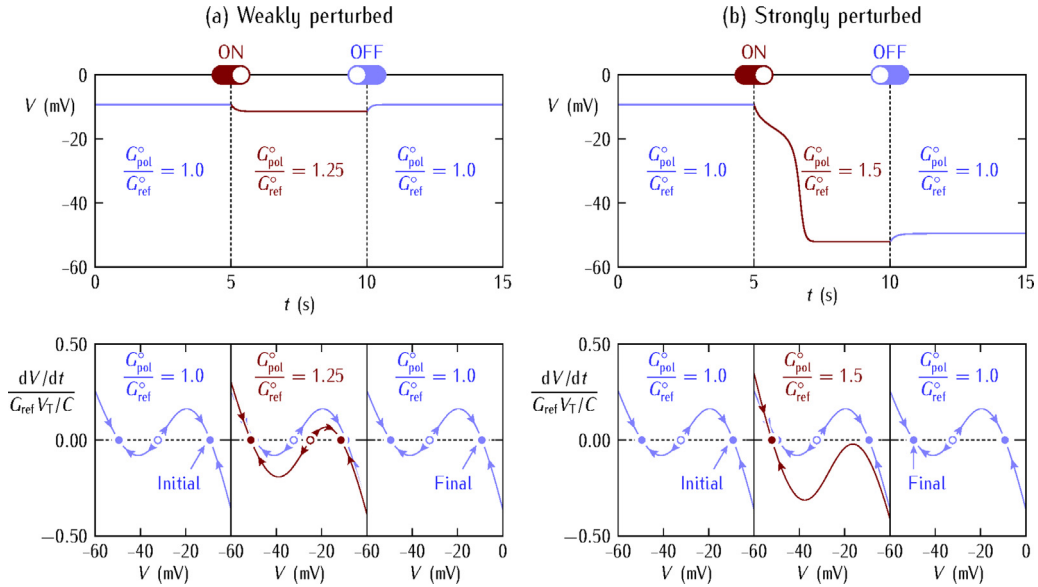


Fig. 6. Cell potential perturbations. (a) A weak perturbation caused by an externally-induced change in the channel maximum conductance ratio G_{pol}^o/G_{dep}^o is not enough to change the cell polarization state. (b) The cell state can however change because of a strong perturbation. The time derivatives of the cell potential, which are given by the channel currents of Fig. 5a, and the resulting potential changes are shown. The transient perturbation may increase the *pol* conductance that promotes the polarized cell state with respect to the *dep* conductance that promotes the depolarized state, which is taken as the reference conductance, $G_{dep}^o = G_{ref}^o$ [19]. In real multicellular systems, these external actions should produce additional diffusion–reaction processes characterized by experimental times much higher than those obtained here assuming immediate conductance changes in the individual cell [15,19,44].

Source: Reprinted figure from Ref. [19].

Interestingly, the electrical activity of neurons can also be modulated by mechanisms that link together ion channel expression and cell potentials [60]. Remarkably, Fig. 7b allows tuning the feedback scheme to each experimental system (Fig. 3) by introducing particular equations for the coupling between transcription and cell potential. We will show different experimental examples of this coupling later.

5. Oscillatory single-cell memory

The feedback between bioelectricity and transcription of Fig. 7 can lead not only to *bi-stable* [22] but also to *oscillatory* [25] cell potentials for the particular case of a negative *pol* channel regulation and a positive *dep* channel regulation (Fig. 8a and b). Eventually, the potential oscillations can be suppressed, and thus one of the two cell states stabilized, by changing the ratio G_{pol}^o/G_{dep}^o of the maximum *pol* and *dep* conductances (Fig. 8b). These conductances, together with the equilibrium potentials E_{dep} and E_{pol} , determine the configuration space for the cell potential V (Fig. 8c). Experimentally, stability diagrams showing bi-stability and transitions between polarized and depolarized states have been found for the case of the Kir channel acting as a sensor and amplifier of neuronal signals in K^+ -mediated neurovascular communication (see Figure 2e of Ref. [53]). Oscillatory potentials and *pol/dep* transitions are also characteristic of the cell cycle [61–65].

In qualitative agreement with the above predictions, Fig. 9 suggests that the coupling among cell potentials and signaling ions (calcium here) can lead to *bi-stable* and *oscillatory* phenomena in different experimental systems. Fig. 9a shows the propagation of an excitation wave as a possible mechanism for signal transmission in pancreatic islets [34]. The time traces of the cell potential and the intracellular calcium concentration are clearly coupled. Fig. 9b schematically shows the relationship between the bioelectric activity, modulated by the cell potential, and calcium levels in *Xenopus* embryogenesis [48]. The methodologies used for *in vivo* studies of developmental calcium signaling allowed the precise positioning of *Xenopus* embryos and it was shown that hyperpolarization and depolarization processes induced dynamic calcium changes at multiple different stages of development [48].

Calcium signaling is essential for cellular life and Ca^{2+} waves can transfer information coded by the amplitude, frequency, and duration of signals [34–38,66]. Thus, cell potentials and signaling ions such as calcium can be experimentally correlated (Fig. 9a) in bioelectrical memories that can encode information within experimental windows of amplitudes and frequencies (Fig. 9b). Note also the short upstream *electrical* times of the excitation wave of Fig. 9a compared with the long downstream *transcriptional* times of Fig. 8b, as discussed elsewhere [8,25].

The extension of the bioelectrical and transcriptional coupling of Fig. 8a, which depends on the signaling ion concentration S , to the case of external electrical stimulations allows to design transcription-control circuits based on

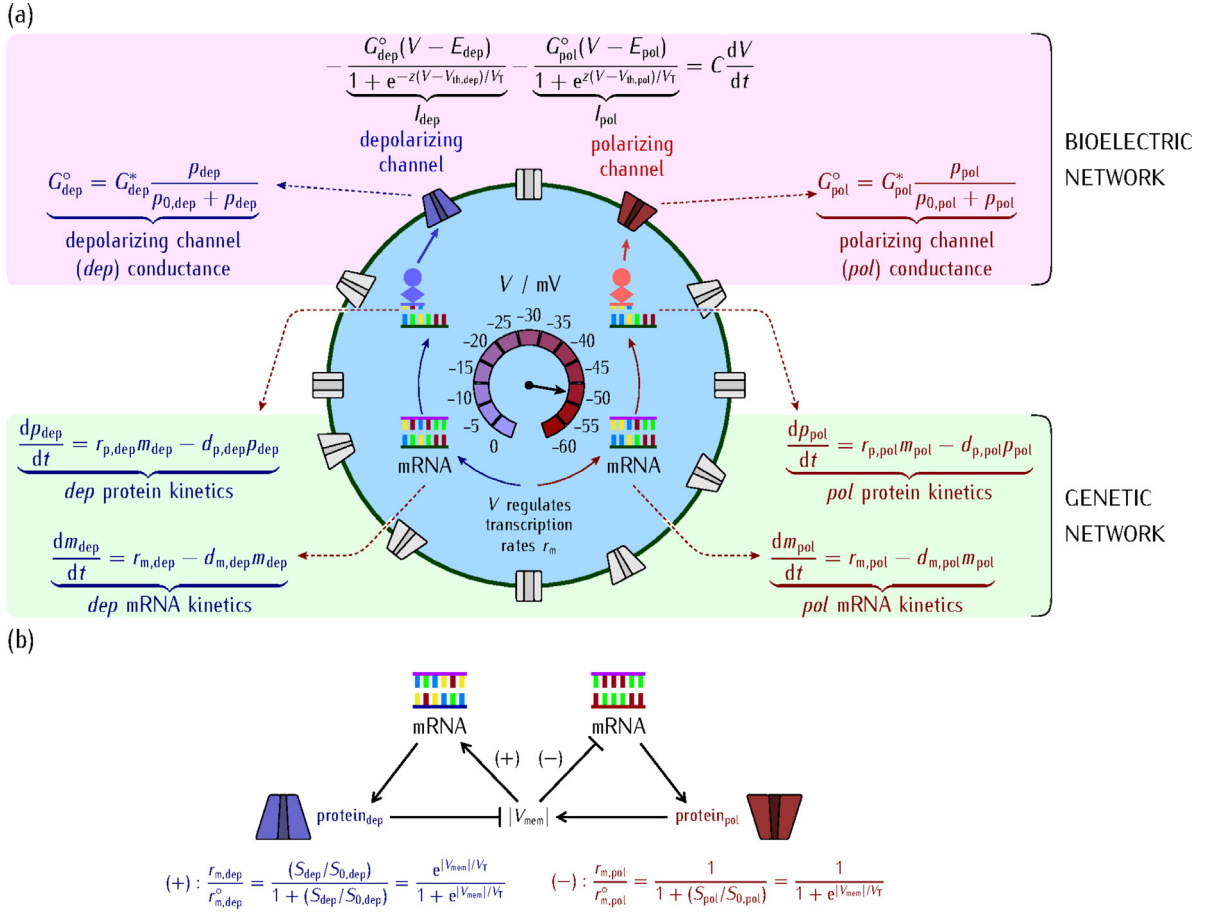


Fig. 7. Coupling of bioelectricity with transcription. (a) The *pol* and *dep* cell states can be modulated by the bioelectrical and transcriptional feedback of the respective channel proteins. (b) This feedback arises here from the voltage-dependent concentration S_k of a signaling ion (e.g. calcium) or molecule (e.g. a neurotransmitter), where $S_{0,k}$ ($k = \text{pol, dep}$) is a reference concentration. In the model, the two ion channels are coupled by the cell potential V because of S_k , which results in transcription rates that may follow a *positive* (+) or *negative* (−) Hill kinetics with respect to V . Note that both the transcriptional and the post-translational dependences of the *pol* and *dep* channel conductances on V are explicitly accounted for in the model. The particular choice of this feedback scheme depends on each specific system. Typical values for all biophysical parameters involved can be found elsewhere [25].

Source: Reprinted figure from Ref. [25].

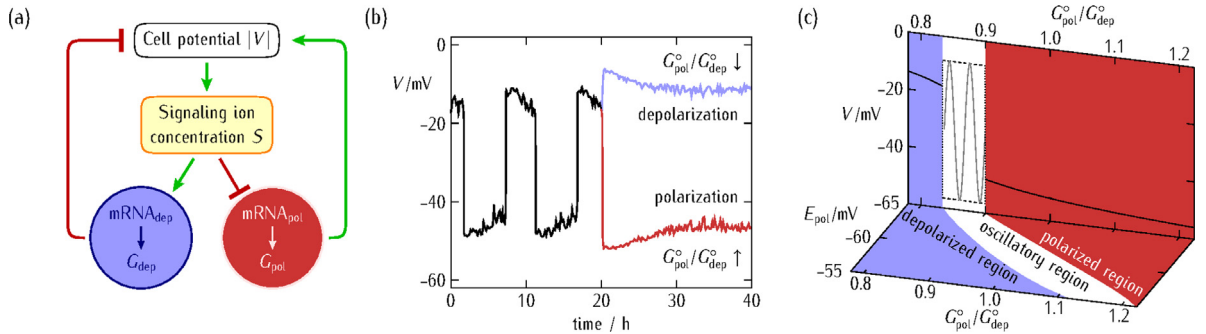


Fig. 8. Bi-stable and oscillatory single-cell cell potentials. (a) The cell potential V characterizing the *pol* and *dep* states, modulated by the bioelectrical and transcriptional feedback of Fig. 7, for the case of a common signaling ion concentration S . (b) The coupling between bioelectricity and transcription leads to oscillatory cell potentials modulated by the maximum conductances ratio G_{pol}^o/G_{dep}^o and the equilibrium potential E_{pol} when the channel protein regulation is negative for the *pol* channel and positive for the *dep* channel. To incorporate the possible noise effects, we assume that the conductances $G_{k,i}^o$ ($k = \text{dep, pol}$) show 5% fluctuations around their respective average values [8]. (c) The conductance ratio G_{pol}^o/G_{dep}^o and equilibrium potential E_{pol} define a configuration space for the bioelectrical transitions between the oscillatory and two stable states.

Source: Reprinted figure from Ref. [25].

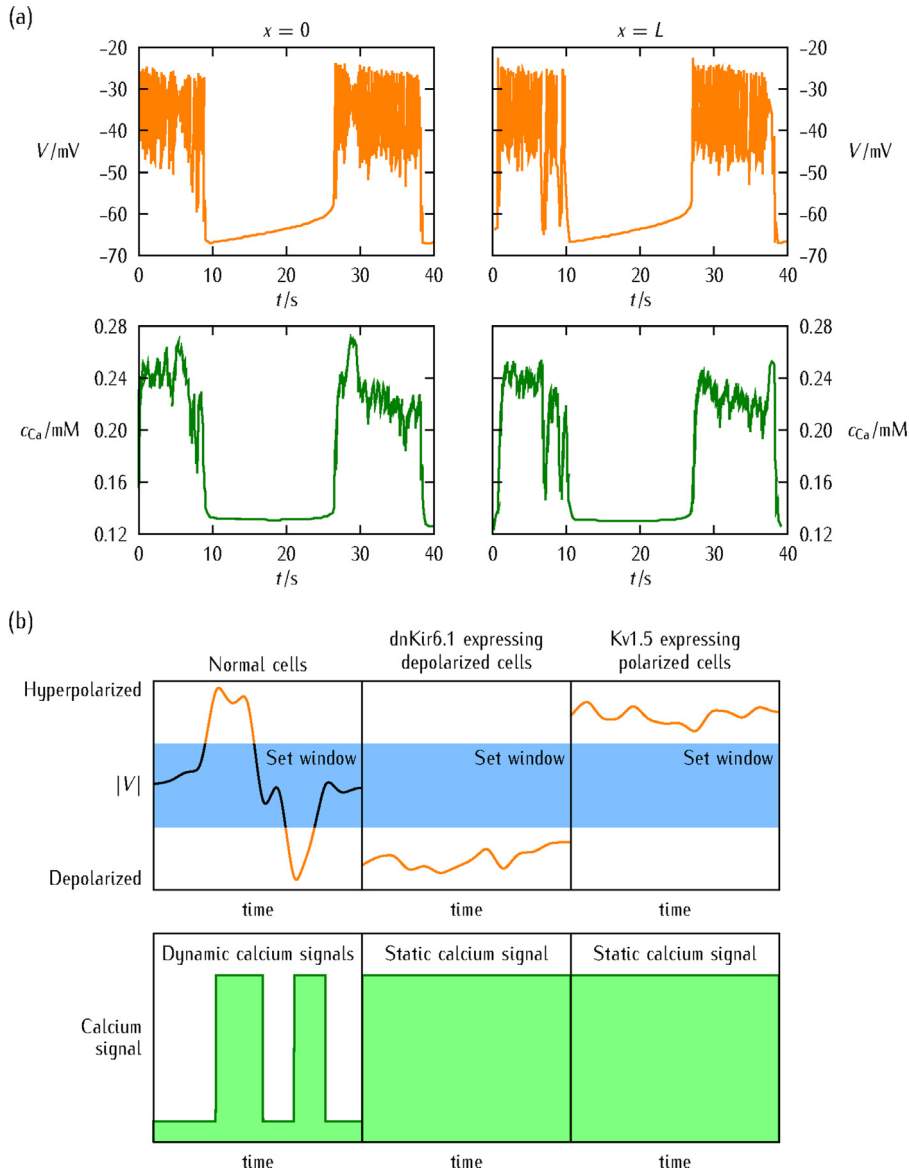


Fig. 9. Coupling between bioelectrical and biochemical signals. (a) The propagation of a bioelectrical excitation wave [34] in pancreatic islets. The activity time delay (vertical dotted lines) for the first ($x = 0$) and last ($x = L$) cells of a linear chain is shown. The correlation between the time traces of the cell potential V and the intracellular calcium concentration c_{Ca} is evident. (b) The absolute values of the cell potential and cytoplasmic calcium levels are also interrelated in *Xenopus* embryogenesis [48], allowing to decode bioelectric events by tracking the interplay between them. Normally developing cells are kept within a target cell potential window. Both cell polarization and depolarization associated with the expression of the two respective exogenous channels (inward-rectifying pore-mutant dnKir6.1 and the hyperpolarizing Kv1.5 potassium channels [48]) cause persistent elevations of cytoplasmic calcium levels that fade over developmental time. These biological examples clearly suggest that ion channels, cell potentials, and signaling ions levels are coupled together in oscillatory and bi-stable memories, in agreement with the model predictions. Source: Reprinted figures from Refs. [34] (Adapted from Biophys J., 80 Aslanidi et al., Excitation wave propagation as a possible mechanism for signal transmission in pancreatic islets of Langerhans, pp. 1195–209 (2001), with permission from Elsevier) and [48].

the cell potential. Fig. 10 shows that an electrical pulse stimulation can provoke the membrane depolarization and subsequent calcium entry by activation of a calcium channel [67]. The authors co-transfected human embryonic kidney (HEK) 293T cells with one of the three L-type voltage-gated calcium channels— $Ca_v1.2$, $Ca_v1.3_{42A}$, or $Ca_v1.3_{\Delta42}$ —encoded by the common subunits and the respective channel-forming subunits shown in the figure. In this case, a voltage-gated calcium channel ($Ca_v1.2$) is coupled to an inwardly rectifying potassium channel ($K_{ir2.1}$) to obtain a voltage-gated circuit that allows to control the expression of a gene of interest in HEK cells. The calcium influx activates a series of downstream processes that eventually trigger this transgene expression, showing again *transcription-control circuits based on cell potentials*. Figs. 5, 7 and 8a describe bioelectrical models that are relevant to describe the experimental coupling

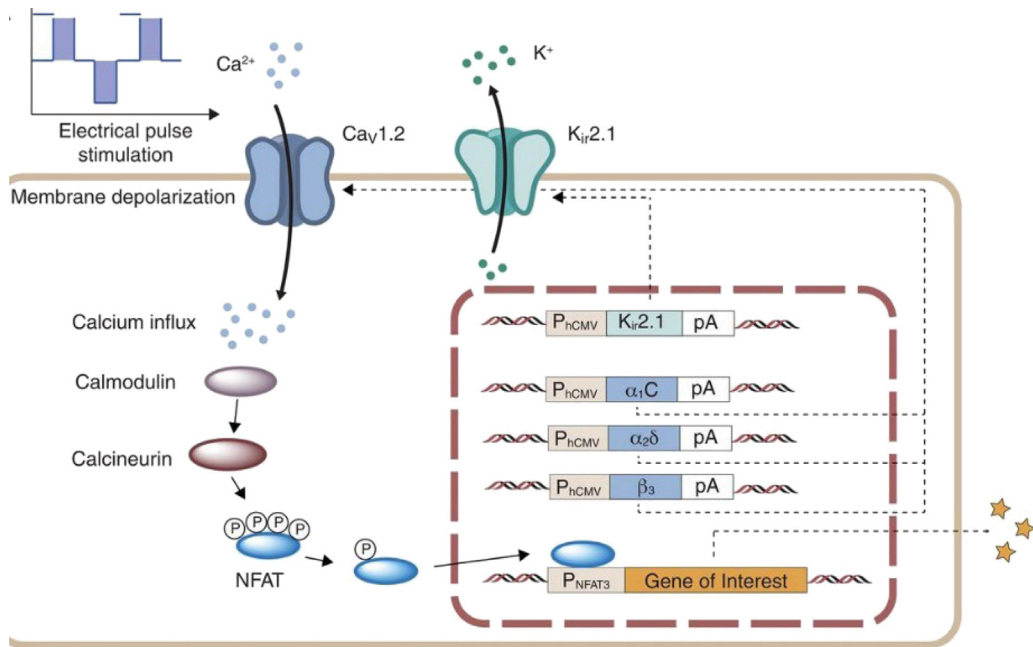


Fig. 10. Cell membrane depolarization-based transcriptional control of mammalian cells. In this electrogenetic circuit design, the inward-rectifying Kir potassium channel controls the membrane potential of the human embryonic kidney (HEK-293T) cells and the electrical pulses depolarize the cell membrane to open the L-type voltage-gated calcium channel (Ca_v). The subsequent calcium entry triggers a series of downstream processes (activation of the calmodulin/calcineurin pathway, dephosphorylation of the nuclear factor of activated T cells (NFAT) and its translocation to the nucleus, activation of the NFAT-sensitive promoter) that eventually lead to the target transgene expression. Note again the coupling between bioelectricity and transcription.

Source: Adapted figure from Ref. [67], K. Krawczyk, S. Xue, P. Buchmann, G. Charpin-El-Hamri, P. Saxena, M.D. Husserr, J. Shao, H. Ye, M. Xie, M. Fussenegger, Electrogenetic cellular insulin release for real-time glycemic control in type 1 diabetic mice, *Science* 368 (2020) 993–1001, with permission from AAAS.

between calcium and potassium channels of opposing rectifying properties (Figs. 5 and 10), together with the relationship between bioelectricity and transcription based on a signaling ion of concentration S (Figs. 7 and 8a).

6. Multicellular pattern memories

We consider now the central question of our study: *how can single-cell potentials be coupled at the multicellular level*. To this end, we will proceed step by step here. We enumerate first the different biological mechanisms that can modulate the cell conductances [8]: (i) a genetic pre-pattern of channels and junction proteins expression, (ii) the post-translational channels and junction blocking by specific ions and molecules, and (iii) the cell potential changes due to the *individual cell* and *intercellular* dynamics. Concerning mechanism (iii), we emphasize that these potential changes may impact on the channels and the junction conductances simultaneously because both can be voltage-gated [8].

The *individual cell* dynamics depend on the relative contributions of the voltage-gated *dep* and *pol* channel conductances G_{dep} and G_{pol} , and have already been considered with detail in the previous section [28]. The *intercellular* dynamics depend on the junction conductance G^0 that characterizes the multicellular connectivity [32] and usually results in collective behaviors that provide tissue-specific cell functionalities, as shown for the case of biochemical signals and networks [58]. As shown previously [8,25], the plasticity provided by the dynamic system connectivity gives *spatio-temporal correlations* between the electrical and transcriptional patterns that are reminiscent of primitive slow neural networks.

6.1. The community effect

We consider first the *community effect* caused by the electric potential (Fig. 11). In the small ensemble composed of a central cell and n neighboring cells, the first cell undergoes a local externally-induced depolarization that decreases the absolute value of its equilibrium potential E_{pol} [19]. The bioelectrical coupling with the neighboring cells, however, eventually forces this abnormally depolarized cell to recover the normal polarized state when the number of surrounding cells is high enough (Fig. 11), consistent with the observation that small regions induced by mRNA injection toward an eye or tumor fate are frequently normalized by their neighbors and prevented from exerting a phenotype [49] –a threshold has to be crossed to establish a stable bioelectric domain. Clearly, n constitutes here a measure of the degree of intercellular

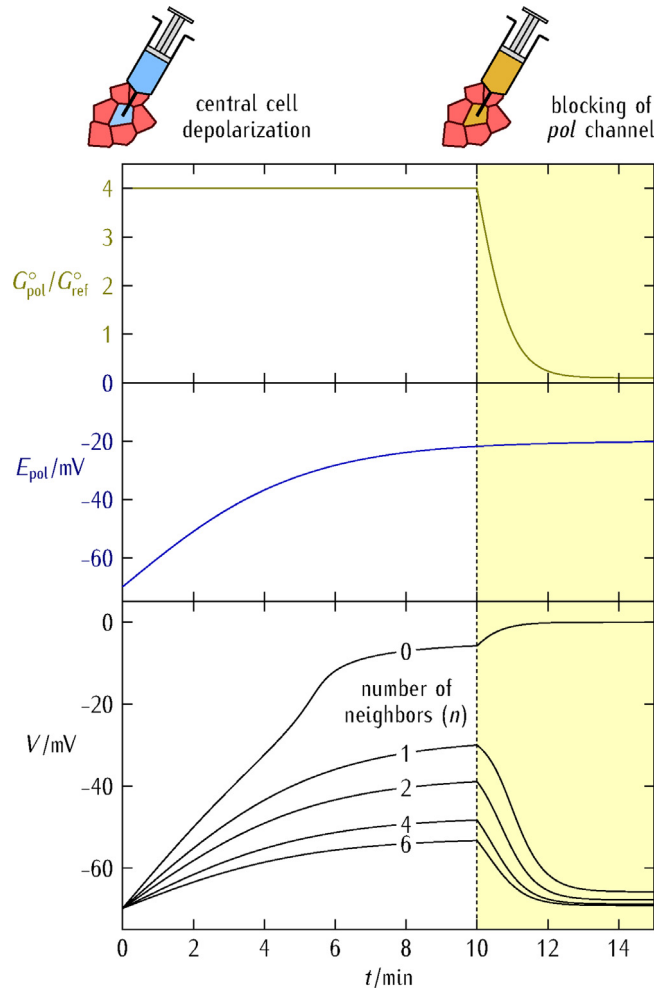


Fig. 11. The multicellular community effect. The depolarization of a central cell can be induced by changing locally the ionic concentration, thus decreasing the absolute value of its equilibrium potential E_{pol} [19], over the time interval $0 < t < 10$ min (center). The resulting decrease in the absolute value of the central cell potential V can be bioelectrically buffered when the number n of intact neighboring cells is high enough (bottom). Later on ($10 < t < 15$ min), we consider the blocking of the cell *pol* channel, simulated here by a slow decrease of its conductance (top). Again, different responses are obtained for the uncoupled ($n=0$) and highly coupled ($n=6$) central cell (bottom). The cell simulation parameters are $G_{pol}^o/G_{ref}^o = 4$ and $G_{dep}^o/G_{ref}^o = 1$ for the channel conductances and $E_{pol} = -70$ mV and $E_{dep} = 0$ mV for the equilibrium potentials. For the sake of simplicity, the gap junction conductances G^o are assumed to be constant (not voltage-gated) in these simulations.

Source: Reprinted figure from Ref. [19].

coupling between the central cell and the neighbors: the uncoupled ($n = 0$) central cell significantly depolarizes, and thus V is close to zero, after the externally-induced depolarization. On the contrary, only a relatively small depolarization is obtained when the central cell is surrounded by a high number of neighbors ($n = 6$) that can buffer its initially polarized potential (Fig. 11).

Fig. 11 also shows the blocking of the *pol* channel of the central cell in the cases of uncoupled and highly-coupled cell. The blocking starts at $t = 10$ min and causes a decrease in the *pol* conductance for $10 < t < 15$ min. The isolated ($n = 0$) central cell depolarizes completely because of the now dominant *dep* channel conductance (Fig. 5b). On the contrary, the central cell can recover the polarized state despite its *pol* conductance decrease when the coupling with the surrounding polarized cells is high ($n = 6$). Again, the average multicellular electric potential acting on the central cell allows a community effect that enforces the isopotential ensemble. Fig. 11 emphasizes the different dynamical responses allowed by the degree of multicellular coupling in response to an external perturbation.

Experimentally, astrocytes are coupled by gap junctions [68,69] that may smooth local changes in the ionic concentrations attempting to depolarize a particular cell. While our model considers only the gap junction currents and does not account for additional effects such as the intercellular potassium transference and buffering, the results obtained can explain qualitatively the experimental trends of Figs. 1i and 3i–g of Ref. [69]. Also, we will show later that buffering the differences across individual cells within a bioelectrical module is central to patterning.

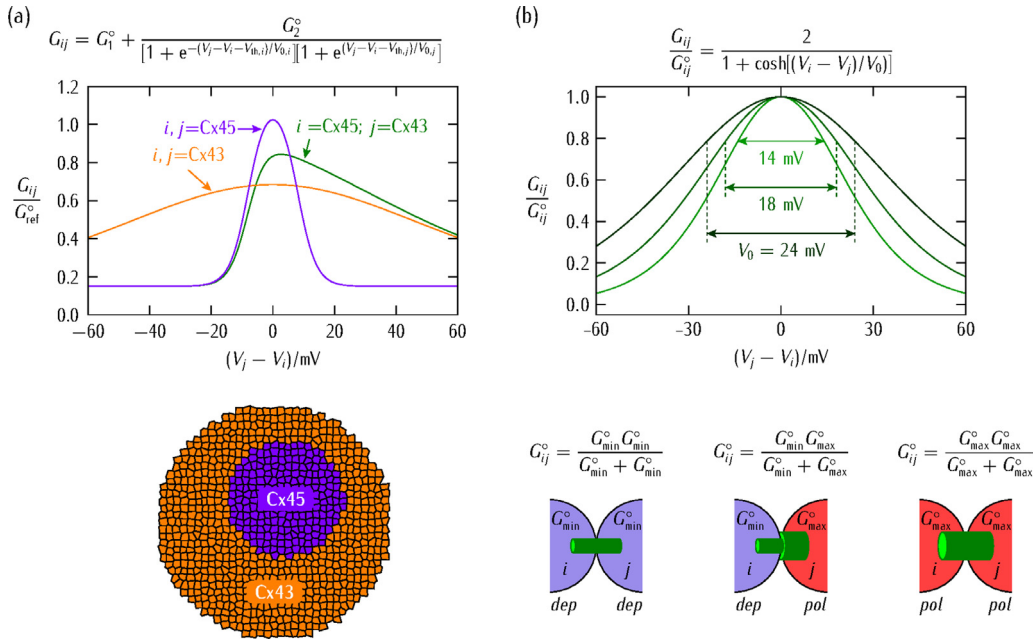


Fig. 12. Intercellular gap junction plasticity. (a) Gap junction conductances (*top*) for two different connexin proteins (*bottom*) and the equation and parameters shown. Symmetrical (homotypic) and asymmetrical (heterotypic) junctions are considered [24]. In the pre-pattern assumed, the inner module expresses connexin Cx45 while the outer module expresses connexin Cx43. At the interface between the two homotypic junction modules, a heterotypic Cx43|Cx45 junction is formed. (b) Two neighboring cells i and j whose maximum individual conductances G_i^o and G_j^o depend on the cell polarization state (*bottom*) should give a maximum intercellular conductance $G_{ij}^o = G_i^o G_j^o / (G_i^o + G_j^o)$. In this case, the expression level of the junction protein is maximum (G_{max}^o) for the *pol* state and minimum (G_{min}^o) for the *dep* state. In addition, the effective gap junction conductance G_{ij} between cells i and j is a bell-shaped function of the potential difference $V_i - V_j$ whose voltage sensitivity depends on the specific junction protein, which is parametrized by the reference potential V_0 (*top*).
Source: Reprinted figures from Refs. [23,24].

6.2. Intercellular gap junction plasticity

Gap junctions are dynamic structures whose intracellular and extracellular modulations allow relatively rapid changes in their number and characteristics at the cell membrane level, thus being crucial components of biophysical plasticity. In addition, they show a myriad of different biochemical and biomechanical functions [70,71] that are far beyond their bioelectrical functions discussed here. We will focus now on their role in the coordination of bioelectrical networks by incorporating a series of experimental facts. In particular, because the intercellular junction conductance is not a passive element and can change with the cell potential for each junction protein type [71,72], we extend the biophysical model to account for three additional effects:

- (i) the junction conductance depends on the protein type (Fig. 12a, *bottom*): different connexin proteins such as Cx43 and Cx46 can show distinct maximum values and voltage-gated behaviors [24,73,74], as shown by type (Fig. 12a, *top*);
- (ii) for a specific connexin, the effective maximum of the junction conductance to be used in the simulations should depend on the expression protein level of each individual cell [23,75], which can be a function of its polarization state determined by the cell potential V (Fig. 12b, *bottom*); and
- (iii) the junction conductance is voltage-gated [23,24,73,74]. The intercellular conductance shows a minimum when one of the cells is either polarized or depolarized relative to its neighboring cell and attains a maximum when the two cells have similar cell potentials (Fig. 12a, *top*, and b, *top*).

Fig. 12a corresponds to mechanisms (i) and (iii) above and suggests opportunities for externally-induced actions if the junction conductance in the *inner bioelectrical module* can be modulated independently of that in the *outer bioelectrical module*. For instance, the relative values of the junction conductances of the two modules can be modified to avoid an abnormal depolarization process in the *inner module* and assist a forced normalization by the *outer module*; different case studies have been described in [24]. In real systems, however, it should be noted that connexins may show other biochemical and biomechanical functions different from those of an electrical conductance.

Fig. 12b corresponds to mechanisms (ii) and (iii) above. As an illustration of these two effects, Fig. 13a shows that if the polarized cell state favors the junction protein expression [23,31], as shown in Fig. 12b (*bottom*), the community effect caused by the depolarized *outer module* on the polarized *inner module* is effective only for very small central patches. On the contrary, the depolarized *inner module* cannot resist the forced polarization by the *outer module* irrespective of

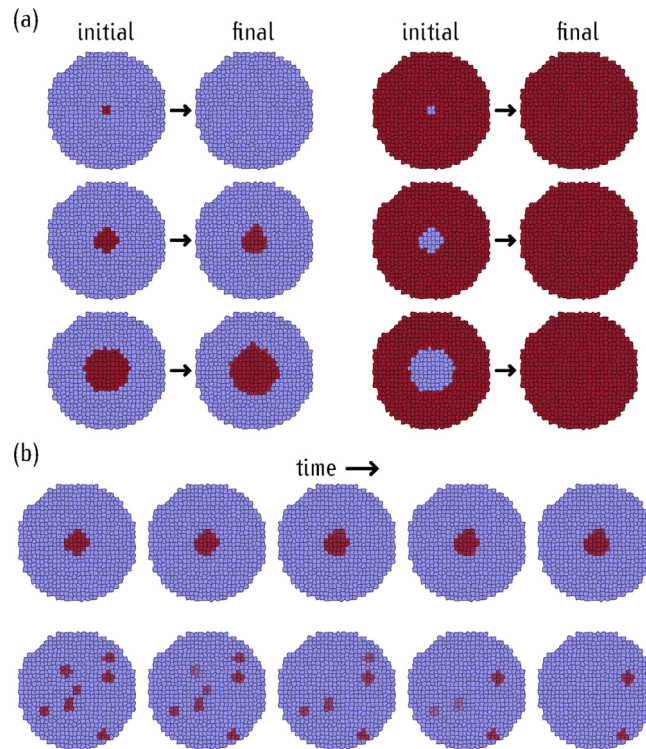


Fig. 13. Cell polarization transitions in multicellular aggregates. (a) The transition from the initial to the stationary state of a multicellular aggregate for different polarized (red) and depolarized (blue) central patch sizes. The dominance of the cell polarized state over the depolarized one is due to the higher junction protein expression assumed in the simulations. The central patch acts here as an *inner bioelectrical module* opposing to the *outer module* actions. (b) The time evolution of the system for the cases of a unique central patch (top) and multiple small patches (bottom) is also shown. Source: Reprinted figure from Ref. [23].

the patch size (Fig. 13a). This result shows that the different junction protein expression of polarized and depolarized cells allows establishing a *symmetry-breaking* effect that can be relevant for multicellular modules in distinct bioelectrical states [22–24]. Fig. 13b suggests also that the number of cells in small modules and their particular location are crucial for the system evolution. While the number of polarized cells in the top panel of Fig. 13b is actually lower than that in the bottom panel, the community effect caused by the depolarized bulk is more significant in the second case.

The single-cell potential influences and is in turn influenced by the membrane voltage-gated ion channels (Figs. 4–7). However, the above examples clearly show that this fact is complicated further by the intercellular coupling provided by the gap junctions (Fig. 14, top). As a complementary view to biochemical concepts and methods, we suggest here that this intercellular coupling can also be important from a bioelectrical perspective, in particular in those cases where different connexins and junctions may coexist in a multicellular environment (Figs. 12 and 14, bottom). Note here that the bioelectrical model described can also account for spatially heterogeneous distributions of different intercellular junctions (Figs. 12a and 13).

Taking together, the results of Figs. 12–14 suggest that the intercellular junctions should support the *integration-segregation* processes needed for the *regionalization of multicellular aggregates* into bioelectrical modules, an experimental characteristic of *pattern memories* (Fig. 1). Note the role of electric potential as a *distributed controller* in the model: (i) the individual cell potential couples the different protein expressions at the *single-cell* level of Fig. 7 and (ii) the multicellular cell potentials couple these expressions at the *module* level because of the intercellular coupling of Fig. 12. Due to this *distributed feedback* between the system bioelectrical state and transcriptional dynamics, the remodeling of the intercellular connectivity can establish different *pattern memories* in development and regeneration [8,9,76].

In the case of tumorigenesis, the fact is that cells can modify their phenotype influenced by complex cell-cell communication processes and intercellular gap junctions composed of specific connexin proteins play also a role here. For instance, cancer stem cells (CSCs) express connexin Cx46, while non-CSCs predominantly express Cx43 [75] so that during differentiation Cx46 is reduced while Cx43 is increased (Fig. 14, bottom). Remarkably, this difference between Cx46 and Cx43 may have also bioelectrical effects because it results in enhanced intercellular communication and decreased cell potential in the case of CSCs [75]. The fact that some gap junctions can be tumor suppressive (Cx43) while others (Cx46) are associated with glioblastoma cancer stem cells (CSCs) networks [75] suggests the targeting of specific gap junctions as a possible clinically relevant procedure to reduce tumor growth. However, it must be mentioned that the junction

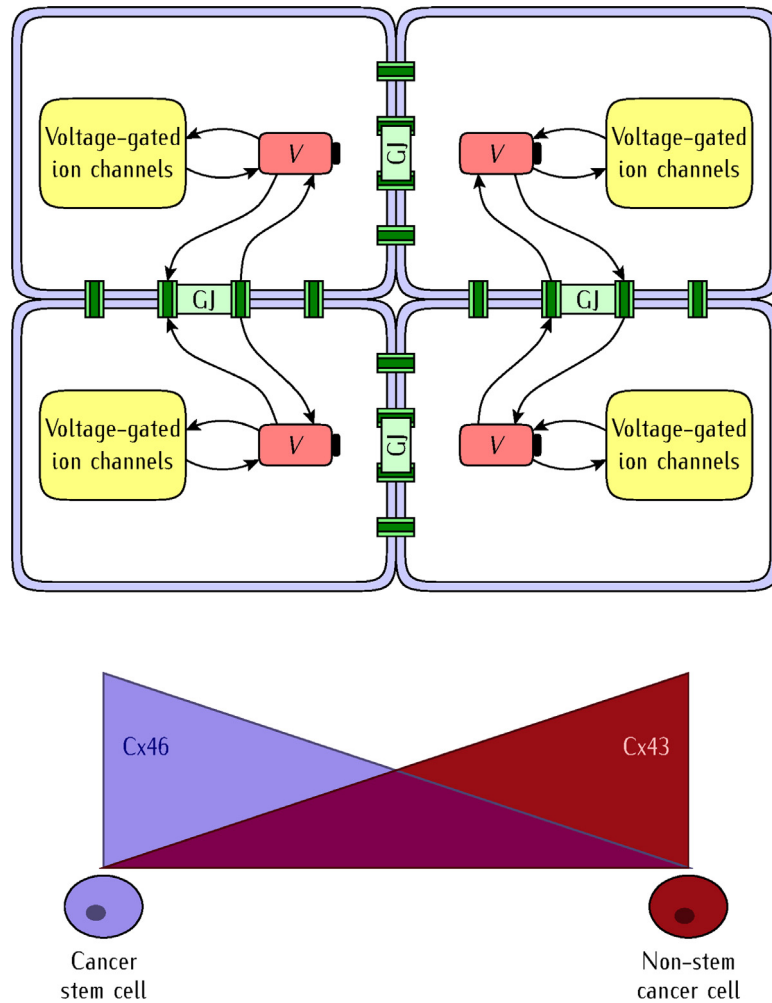


Fig. 14. Single-cell potentials (V) influence and are influenced by voltage-gated ion channels. In turn, this feedback is coupled by the intercellular gap junctions (GJ) at the multicellular level (top). The arrows indicate the feedback processes. Note also that different connexins and junctions may coexist in a multicellular environment. For instance, glioblastoma cancer stem cells show different gap junction proteins that influence self-renewal and tumor growth (bottom). In particular, connexin 46 (Cx46) has been identified as a novel glioblastoma (GBM) cancer stem cell regulator so that gap junction targeting may inhibit GBM growth. On the contrary, the gap junctions based on connexin 43 (Cx43) appear to be tumor suppressive in this case [75]. These facts suggest that bioelectrical models should account for spatially heterogeneous distributions of the different intercellular junctions, as shown in Fig. 12 [24].

Source: Adapted figures from Refs. [19] (top) and [75] (bottom).

tumorigenic role is context-dependent and depends both on the particular connexin involved and the tumor stage (see Ref. [24] and references therein). In addition, the junctional role of connexin proteins is not limited to transducers of electrical signals because they can also transfer biochemical signals, as we will consider later, and show also channel-independent functions with pro- and anti-tumorigenic effects. Thus, it may constitute an oversimplification to focus only on the bioelectrical aspects of gap junctions in real problems.

6.3. Bioelectrical pattern memories: examples

We consider now different case studies that emphasize further the concepts of *bioelectrical module* and *pattern memory*. A previous question here is how an initial pre-pattern can be established across the system. Spatial patterns can emerge from the dynamics of relatively simple bioelectrical models and do not require complex pre-patterns of protein expression. In fact, these patterns may arise from symmetry-breaking and amplification phenomena in quasi-homogeneous multicellular systems because of morphogen gradients, local electrical and mechanical fields, and the different junction dynamics of polarized and depolarized cells [8–10,24,55,56,77–81]. We will not address this problem here in detail and instead focus our study on the bioelectrical characteristics of different pattern memories.

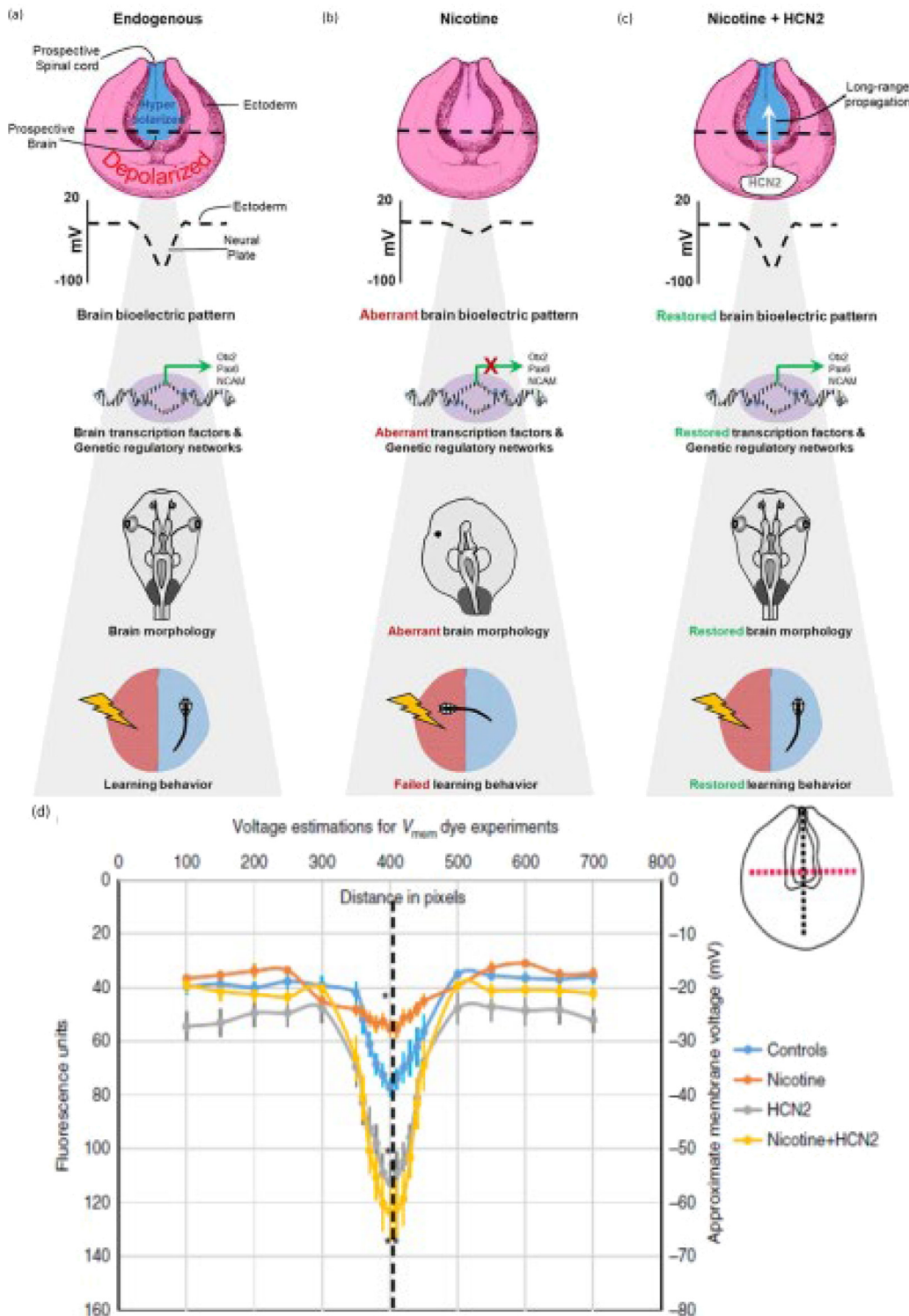


Fig. 15. Brain teratogenesis and bioelectrically-induced long-range repair [15]. (a) Scheme emphasizing the multicellular potential pattern in the neural plate and surrounding ectoderm (top). These two regions act here as *bioelectrical modules* that show *internal integration* but *external segregation* between them. The original multicellular pattern behaves as a *bioelectrical memory* crucial for gene expression, brain morphology, and learning (bottom). (b) Embryonic exposure to nicotine erases this memory pattern and leads to aberrant gene expression, brain morphology defects, and impaired learning. (c) Transplanting a *new bioelectrical module*, the HCN2 channel tissue, onto nicotine-exposed embryos restores the correct memory pattern and rescues the correct gene expression. (d) The voltage reporter dyes show the effect of the HCN2 channel on the different control and nicotine-exposed electric potential patterns [77]. The patterns are obtained along the red dotted line in the inset and correspond to stage ~15 *Xenopus* embryos.

Source: Reprinted figure from Refs. [15,77].

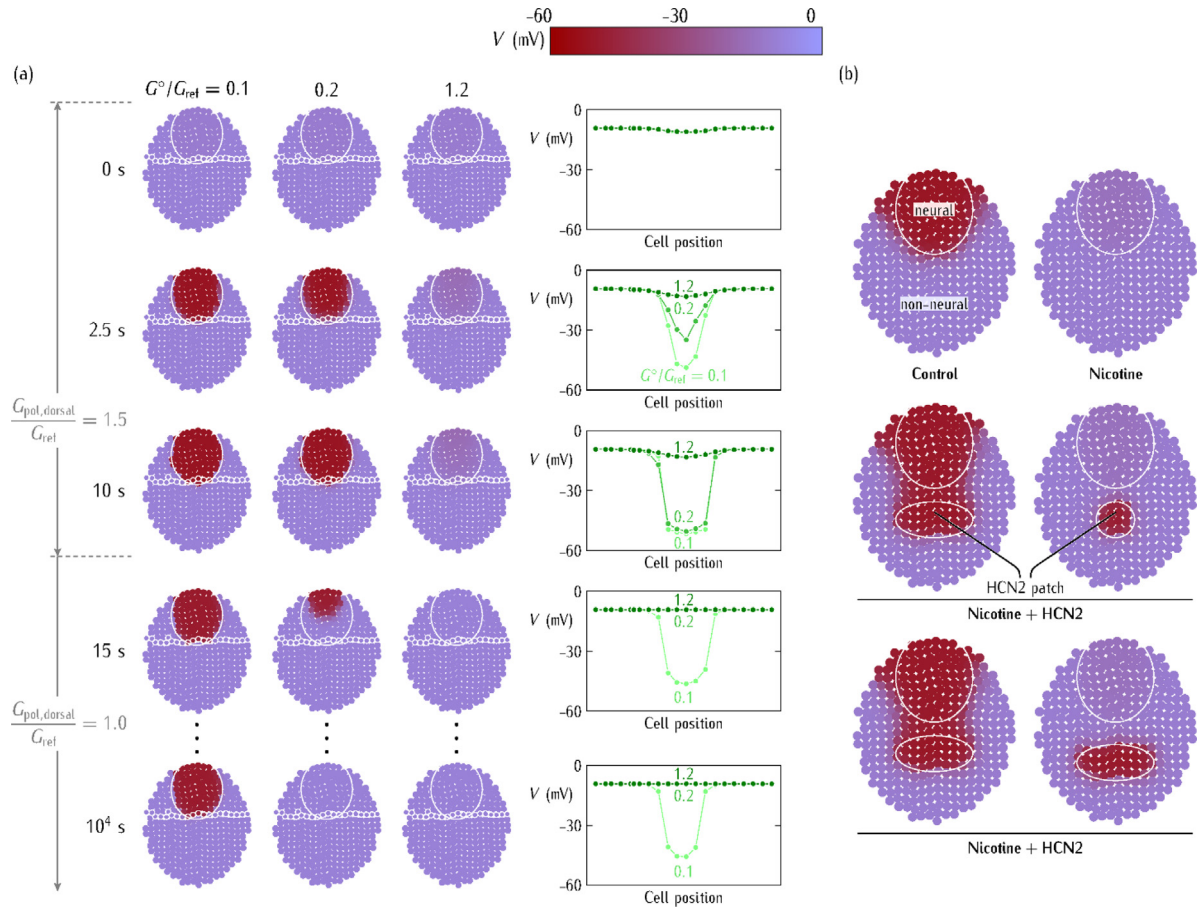


Fig. 16. Simulations of bioelectrically-induced long-range repair [15]. (a) Spatio-temporal potential patterns for the experiments of Fig. 15 showing the two relevant bioelectrical modules in a minimal model. The effect of the insertion of a polarizing channel and the resulting subsequent repolarization is simulated here by increasing the *pol* channel conductance in the initially depolarized dorsal region [19]. Even when the transient stimulus is retired, the cells in the dorsal region can keep the polarized state for low enough intercellular conductances $G^{\text{pol}}/G_{\text{ref}}$ (Fig. 6). The line graphs show the electric potential patterns for the cells along the ventral–dorsal–ventral regions for different $G^{\text{pol}}/G_{\text{ref}}$ values. (b) The model results concerning the HCN2 tissue experimental transplantation of Fig. 15 [15]. The simulations qualitatively predict this *distant* channel-mediated rescue of the bioelectrical memory pattern in nicotine-exposed embryos under different conditions concerning the size and position of the inserted channel tissue. Source: Reprinted figures from Refs. [15,19].

Fig. 15 shows a tentative model for bioelectric signaling in brain teratogenesis and long-range repair in *Xenopus* frog [15,45,77]. The electric potential difference between two bioelectrical modules, the normally polarized neural plate and the depolarized neighboring ectoderm, allows for correct gene expression pattern, brain morphology, and learning. However, embryonic nicotine exposure tends to dissipate this pattern memory of modules, which leads to aberrant gene expression pattern, brain morphology defects, and impaired learning. The transplantation of a new bioelectric module composed of a HCN2 channel tissue onto nicotine-exposed embryos restores the correct multicellular potential pattern, thus rescuing gene expression pattern, brain morphology defects, and learning in the animals. The model simulations of Fig. 16 offer qualitative insights into the bioelectricity of the system.

The bioelectrical action at distance of Fig. 16b could also be involved in ectopically-induced eye lenses on tadpole tails [16,49]. In this case, cells expressing a specific ion channel can recruit neighboring, genetically normal, cells not having this channel to collaborate in the new lens by establishing a common cell potentials window [49]. The simulations of Figs. 5, 6, and 8 suggest that combinations of channels with different voltage-gated conductances allow to establish potential windows while those of Fig. 16 show that a local bioelectrical perturbation can propagate in the multicellular aggregate, as experimentally demonstrated [15,56]. Other options to establish potential windows are the concerted action of calcium voltage-gated channels [48] and the salt-induced modification of the *dep* and *pol* equilibrium potentials E_{dep} and E_{pol} in the model, which depend on the potassium and sodium concentrations [49].

Because bioelectrical, transcriptional, and diffusional times can be very different [8,22,44], the above external actions will give slow diffusion-reaction processes rather the immediate channel conductance changes assumed in the simulations. This fact will result in experimental times much higher than those obtained here with a purely electrical approach

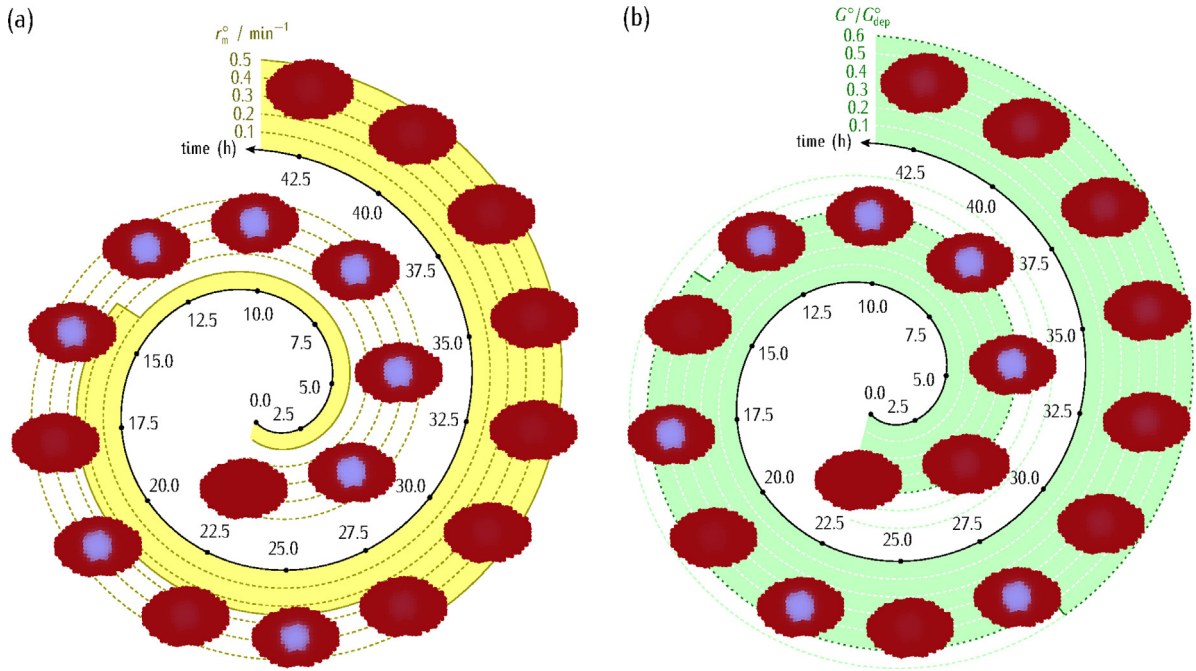


Fig. 17. Bi-stable and oscillatory multicellular pattern memories. (a) Dynamic changes in the single-cell *pol* channel expression of the inner bioelectrical module can establish bi-stable and oscillatory multicellular pattern memories [76]. At time $t = 0$, $N_p = 55$ cells in this module, located within an aggregate of $N = 304$ cells, are in a polarized state that is not stable (Fig. 5) because of the locally low values of the *pol* channel protein transcription rate r_m^o . The subsequent snapshots describe the sequence of depolarized–oscillatory–polarized states of the module when r_m^o is increased from 0.1 to 0.5 min^{-1} . (b) Alternatively, the remodeling of the intercellular connectivity can also allow similar transitions in the module multicellular states when the junction conductance is increased following a series of steps.

Source: Reprinted figure from Ref. [76].

and a relatively small number of model cells. However, our point here is to show how the spatially separated *bioelectrical modules*, defined by the system *integration-segregation topology*, provide a morphologically instructive pattern memory in a model multicellular aggregate, not to describe quantitatively the real system. A more complete model that combines bioelectricity and biochemical networks is described in [77].

Thus, while it is experimentally feasible (Fig. 1) that the above bioelectrical patterns influence downstream diffusional and transcriptional processes [8,9,12,46,82,83], other biomechanical and biochemical signals not included here should also be important [10,40,84,85]. For instance, signaling ions and electric potential gradients can be coupled together [22,44] and the diffusion of neurotransmitters can be regulated by voltage-dependent transporters [9]. In this case, not only the electric potential gradient but also the electrophoretic transport of chemical morphogens following this gradient should be incorporated in the model. The above simulations suggest only that a binary information based on the bioelectrical modulation of polarity may influence subsequent developmental programs [10].

6.4. Bi-stable and oscillatory pattern memories

The comparison between Figs. 8b and 9a immediately suggest that oscillatory coupling between bioelectrical and biochemical signals can occur in biological systems. We next consider the consequences of this coupling in multicellular patterns. Experimentally, electrical synchronization in distant non-excitable cells via gap junction networks can be involved in the activation of downstream processes in embryogenesis, regeneration, and healing [8–10,43].

Using the model of Figs. 7a and 12b, the simulations of Fig. 17 show how local changes in the expressions of channels and intercellular junction proteins can implement *bi-stable* and *oscillatory* multicellular pattern memories. In these illustrative examples, we assume that the *pol* channel transcription rate (Fig. 17a) and the effective junction conductance (Fig. 17b) increase with time. The different bioelectrical pattern memories, which are retrieved for particular *input* values of the system bioelectrical parameters, can provide local binary and oscillatory information for downstream transcriptional processes, as suggested by the electrical patterns of Fig. 1. In this way, two adjacent bioelectrical modules consisting of cells in different polarization states could stabilize distinct transcriptions and tissue types. In addition, these modules can be reprogrammed, changing the respective bioelectrical and transcriptional states of Fig. 7a as a function of the intercellular coupling of Fig. 12, thus leading to the transitions shown in Fig. 17b. The transitions show that not only

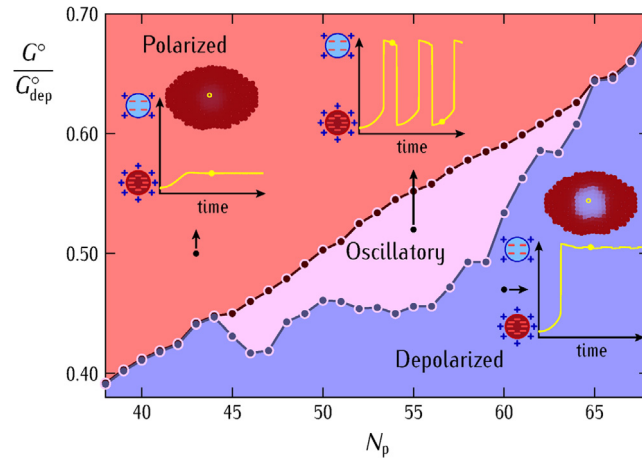


Fig. 18. The inherent capacity of a multicellular system to adaptively move in bioelectrical configuration spaces. The individual cell states are shown as a function of the number N_p of cells in the inner module and the ratio of the junction conductance to the dep channel conductance, G^o/G_{dep}^o [76]. The insets give the time evolutions of the potential for a central cell in the module. Oscillatory phenomena are possible for intermediate values of the intercellular connectivity and the number of cells. The module can transit to new bioelectrical states at certain size and connectivity values, thus retrieving pattern memories that may have morphological consequences [8,9].

Source: Reprinted figure from Ref. [76].

the steady-state distributions of absolute cell potentials within a module but also the time-dependent relative potentials between adjacent modules can be significant for the instructive patterns (Fig. 1).

Fig. 6 showed that transient manipulation of target ion channels can lead to re-settings of cell bioelectrical states. Fig. 17 suggest that this result can be extended to the multicellular case by appropriate external actions on ion channels and gap junctions. Importantly, Fig. 17 shows that similar bioelectrical patterns can be established by acting at different system levels: the *individual cell* channel protein transcription (Fig. 17a) and the *intercellular* junction conductance (Fig. 17b). The first effect can be induced by appropriate transcription factors while the second one is obtained by blocker molecules such as octanol [30]. In this context, general anesthetics are considered junction blockers that disrupt intercellular communication reversibly. The correct bioelectrical state can be recovered after the connections are reestablished again, as suggested by moving back the values of the junction conductance G^o/G_{dep}^o in Fig. 17b. Thus, signals that tune the multicellular coupling, e.g. via the junction protein expression, should provoke a cascade of transitions between different self-organized tissue states [24,31,55,56].

To better understand the different bioelectrical states of Fig. 17, Fig. 18 shows the configuration space for the single-cell states in the inner module as a function of the number of cells in this module and the intercellular junction conductance. The time evolutions of a single-cell potential in the stable and oscillatory bioelectrical memories are schematically visualized. Remarkably, the stable *depolarized* state characteristic of each *inner* module cell of Fig. 18 can transit to the *oscillatory* and *polarized* states when forced by the *outer* module cells. These transitions are induced by increasing the multicellular coupling (G^o/G_{dep}^o) when the inner module population is below a critical size (N_p), thus suggesting that the stability of single-cell states is a collective rather than an individual property in multicellular systems. Also, the comparison of Fig. 8c for the configuration space of single-cell states with Fig. 18 for the multicellular module states clearly shows how *bioelectric quorums* can be established by increasing the system connectivity, analogously to increasing communication in bacterial communities [86] where the encoding of a memory can be based on the coupling between the membrane potential and the induced oscillations in extracellular ion concentrations [87].

Because cell bioelectric states are related to differentiation processes (Fig. 1), cell-type proportions can also be modulated not only by biochemical [88] but also by electrical multicellular coupling. Experimentally, this bioelectrical modulation influences gene expression during development and homeostasis [9] as well as wound healing [36]. We will not address these biochemical downstream processes because we will focus here on the description of the bioelectrical modules and the resulting patterns. Note however that the model of Fig. 7 for the coupling between bioelectricity and transcription can readily be extended to other proteins provided that the specific kinetic mechanisms operating in each specific case are known [83,89].

Taken together, Figs. 12–18 show that voltage-gated gap junctions may allow *short range integration* within regions of similar cell potentials (and thus *high internal* junction conductances) but *long range segregation* between regions of different cell potentials (and thus *low interfacial* junction conductances). These separate subsystems constitute the *bioelectrical modules* that support the spatio-temporal *pattern memories* in the model, which are defined by the particular *integration-segregation topology* of the different modules in the multicellular aggregate. This topology is vaguely reminiscent of a multicellular landscape [90,91] with locally different bioelectrical and transcriptional states. While the above simulations emphasize qualitative concepts only, Fig. 1 suggests indeed that the integration-segregation of the different electric

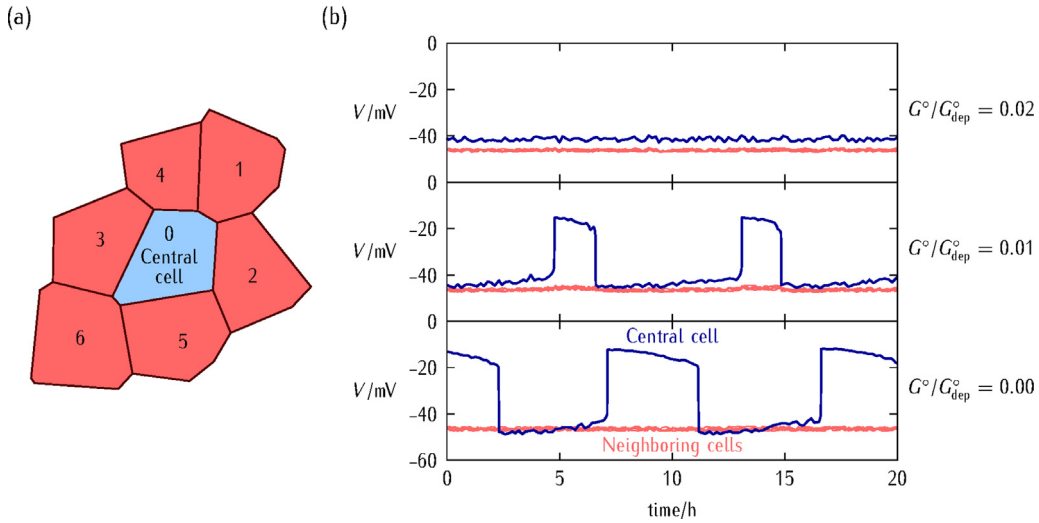


Fig. 19. Oscillatory phenomena and the community effect. (a) A central cell # 0 that suffers a local depolarization and enters the oscillatory regime is brought into contact with neighboring cells #1 – 6 that are in quiescent polarized states. The central cell is connected through gap junctions to the neighboring cells and the model is that of Figs. 7 and 12. (b) The subsequent changes of the central (blue curves) and neighboring (orange curves) cells potentials are shown for three intercellular junction to dep channel relative conductances G^o/G^o_{dep} . For the zero intercellular conductance case, the central cell is isolated from the rest of cells and thus can keep oscillating. On the contrary, the neighboring cells enforce the quiescent polarized state to the central cell for moderate intercellular conductances. The potential fluctuations are simulated by a white noise of amplitude 0.06 mV and characteristic time around 10 min which is applied to the equilibrium potential E_{pol} .

potential patterns allows the combination of plasticity and robustness characteristic of morphology, the establishment of local organs, and tumorigenesis [9].

6.5. Oscillatory phenomena and synchronization

We study now the case of oscillatory bioelectrical phenomena in multicellular aggregates. Fig. 9a shows that there is a coupling between the electric potential and the calcium level in pancreatic islets (34). Interestingly, there exists also an interplay between the cell potential and the different phases of the cell cycle [61–64]. The oscillatory potentials along the cell cycle are schematically described e.g. in Fig. 1 of Ref. [61] and Fig. 1 of Ref. [65].

We consider first the case of a central cell surrounded by quiescent cells in the polarized stable state of Fig. 5b. Let us assume that this central cell suffers a local depolarization and enters into the oscillatory regime of Fig. 8b, thus progressing through the cell cycle. On purely electrical grounds, it could be anticipated that the intercellular junction conductances would counteract this local cell depolarization because the neighboring cells are in the polarized state, as shown in Fig. 19a. However, this bioelectrical buffering, which is due to the community effect of Figs. 11 and 13, could not occur for an uncoupled central cell. Thus, this cell can progress through the cycle in the zero junction conductance case of Fig. 19b.

Fig. 19 suggests that the loss of intercellular coupling may allow a transiently depolarized central cell to escape from the quiescent polarized state imposed by the neighboring cells and eventually achieve an oscillatory potential regime, which is one of the bioelectrical characteristics of the cell cycle [61]. Experimentally, defective intercellular communication caused by non-functional gap junctions is related to the initial states of uncontrolled growth [92], although the particular junction role can vary along the different tumorigenesis stages and is still under discussion [24,93]. In addition, the intercellular junctions allow not only the transfer of electrical currents (Fig. 11) but also the exchange of biochemical signals [44,92,93], which is ignored here. Interestingly, the community effect of Fig. 19 could not be effective for a large enough patch of depolarized cells [22], which may be relevant to establish separate bioelectrical modules in development and regeneration, as shown in Figs. 1, 16, and 17.

We will also explore the possibility of *synchronization* phenomena in multicellular aggregates. We have shown previously that the transcription and degradation rates of the model equations (Fig. 7) can define single-cell frequencies that couple together bioelectrical potentials and channel protein concentrations [25]. Fig. 20 shows that cell groups with different individual frequencies around two average separated values can eventually collapse to a synchronized multicellular aggregate at high intercellular junction conductances. This result is reminiscent to oscillatory phenomena observed across different experimental systems: biological rhythms and patterns in temporal self-organization [94–97], noise and excitability phenomena in cancer cells aggregates [98,99], and bioelectrical waves in pancreatic islets [34,100] and bacterial communities [86,87]. Note in particular that increasing the intercellular communication here may be related to enhancing the density and quorum sensing effects in bacterial communities [86,87] and spatially-structured oscillatory

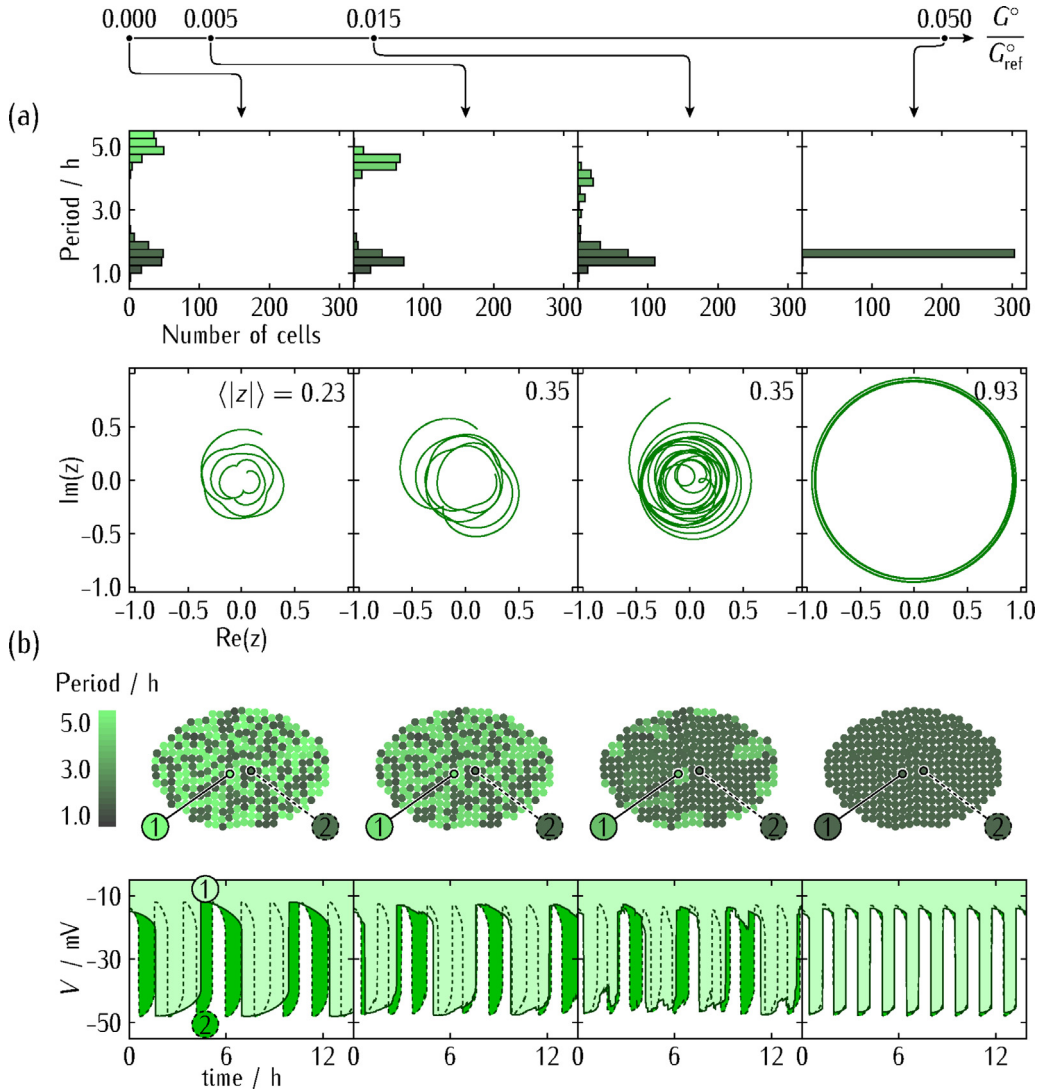


Fig. 20. Oscillatory phenomena and synchronization. (a) A spatially random bimodal distribution of single-cell frequencies synchronizes into one multicellular frequency by increasing the intercellular coupling, where G^o/G_{ref} is the junction conductance scaled to a reference conductance. The multicellular aggregate trajectories are characterized in terms of the order parameter z in the complex plane [25]. (b) The different individual cell frequencies eventually merge into a synchronized aggregate, as shown by two particular cell potentials.

Source: Reprinted figure from Ref. [25].

cell populations [101,102]. In addition, because of the interplay between the cell potential and progression through the cycle [61,63,64], the multicellular synchronization of electric potentials might influence the collective regulation of individual cell cycles. Synchronization between neural networks has also been proposed as a possible mechanism for information transfer modulated by the gap junction plasticity [103].

Note the wide range of spatial scales where oscillatory phenomena occur: the *single-cell* case of Fig. 8, the *small inner patch* of Fig. 17, and the *whole system* of Fig. 20. Individual cell oscillations can be regulated both at the *bioelectrical module* and *multicellular aggregate* levels because of the integration-segregation phenomena provided by the local microenvironment and the intercellular junctions, which should be useful for electrical stimulation and recording across multiple spatial and temporal scales in bioelectrical interfaces [104].

Fig. 21 considers the case of locally coupled and uncoupled multicellular modules [19]. Again, the oscillatory cell potentials result from spatially-random single-cell frequencies following a bimodal distribution (Fig. 21a). The bioelectrical synchronization/desynchronization of the central cells shows that multicellular patterns can be encoded both at the spatial and temporal levels exploiting the system connectivity (Fig. 21b). Remarkably, the coherent oscillations of Figs. 20 and 21 occur with no central controller.

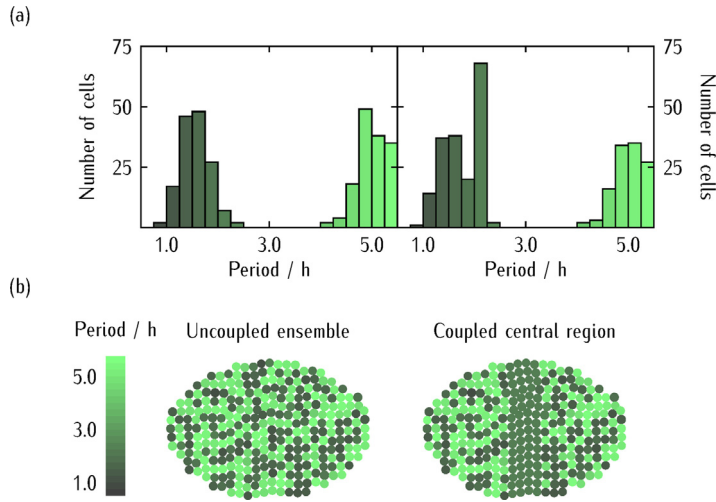


Fig. 21. Oscillatory phenomena and synchronization with local coupling. (a) Bimodal distributions of single-cell frequencies. (b) A central module of coupled cells with a relative junction conductance $G^0/G_{\text{ref}} = 0.5$ and two neighboring uncoupled modules with $G^0/G_{\text{ref}} = 0$. Note that no synchronization is possible if all system cells are uncoupled. Transitions between the patterns can be triggered by a dynamic system connectivity. Source: Reprinted figure from Ref. [19].

The specific system connectivity assumed in Fig. 21 can result from spatially different junction protein expressions (Fig. 12a). It is tempting to speculate that the respective patterns showing bioelectrical desynchronization and synchronization might constitute a slow-wave reminiscent of the fast coupling and uncoupling phenomena observed in distant brain modules [105]. It should be noted however that the mechanisms involved in brain large scale integration are far more complex than those considered here and, in particular, they may use multiple frequency bands. In our case, the dynamic connection and disconnection of functional bioelectrical modules as a function of environmental signals should be relevant for the time-dependent experimental patterns of Fig. 1 because the synchronization/desynchronization of particular modules in the system can assist the experimentally observed segregation–integration phenomena; see the different connectivity maps of Fig. 21b in the light of the time sequence of modularities shown in Fig. 17b.

Although the above ideas are qualitatively suggestive, the fact is that the interplay between membrane electric activity and gene expression in biological clocks involves a multitude of signaling pathways not considered here. With this limitation in mind, multicellular aggregates of non-excitable cells could be regarded as a primitive form of neural networks where the synaptic proteins contribute to memory consolidation via synchronization and oscillatory coherence [106–108]. Similarly, the intercellular junction proteins of Figs. 12 and 13 act as voltage-dependent conductances that change dynamically with the states of the neighboring cells, thus allowing *pattern memories* composed of *bioelectrical modules* arranged in different topologies. Also, the channels and junction proteins allow the correlation between transcription and bioelectricity in these pattern memories, much as the synaptic coupling between biochemical and electrical signals allows activity-dependent transcriptional processes in information processing and memory [107]. We believe that this analogy between different forms of multicellular information processing supports further our proposal of bioelectrical pattern memories as local regulators of transcription.

6.6. Extensions of bioelectrical model

In this section, we propose possible extensions of ion channel-based bioelectrical models to describe the cases of diffusible signaling molecules [22,44,85], cell potential-mediated cellular binding of charged nanoparticles to membranes [22,109,110], and coupled biomechanical and bioelectrical effects in cell membranes [111,112].

We have considered the reaction–diffusion of microenvironmental channel blockers [22] and the intercellular transfer of microRNA (miRNA) through voltage-gated gap junctions [44]. In the second case, note that the junctional role of connexin proteins is not limited to transducers of electrical signals because they transfer also biochemical signals. Thus, it is of biophysical interest to consider the transport of these signals through intercellular gap junctions. Figs. 22 and 23 show extended single-cell and intercellular models, respectively, that include now the spatiotemporal miRNA, mRNA, and channel protein concentrations [44]. The pattern memories obtained suggest that the miRNA intercellular transfer can also permit the modulation of the single-cell transcriptional and bioelectric states at the multicellular level because of average electric potentials.

The biophysical simulations presented here can also be extended to other experimental problems where diffusion–reaction phenomena coexist with gap junction blocking [22,85]. In particular, the dynamics of the intercellular gap junction network can be associated with the spatial patterning of differentiation and multicellular morphology [30,31].

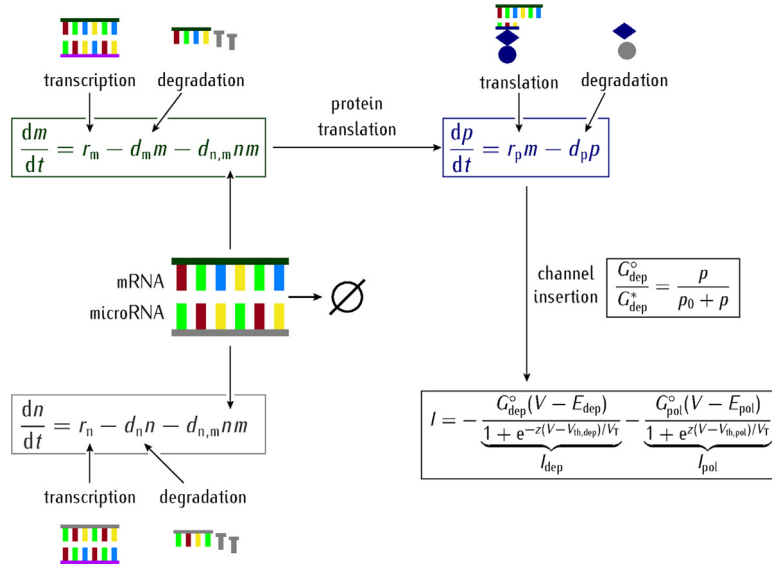


Fig. 22. Extension of the model of Fig. 7 for the transcriptional and bioelectrical feedback at the single-cell level in the case of miRNA [44,89]. For the sake of simplicity, the model assumes that the cell potential V regulates only the protein concentration p of the depolarizing channel conductance G_{dep}^o . Note that the kinetic equations for the miRNA (n), mRNA (m), and protein (p) concentrations include the transcription rates of miRNA (r_n) and mRNA (r_m) together with the translation rate of protein (r_p) and the degradation rates d_n , d_m , and d_p . Also, the miRNA-mRNA irreversible association is described by the degradation rate $d_{n,m}$.

Source: Reprinted figure from Ref. [44]

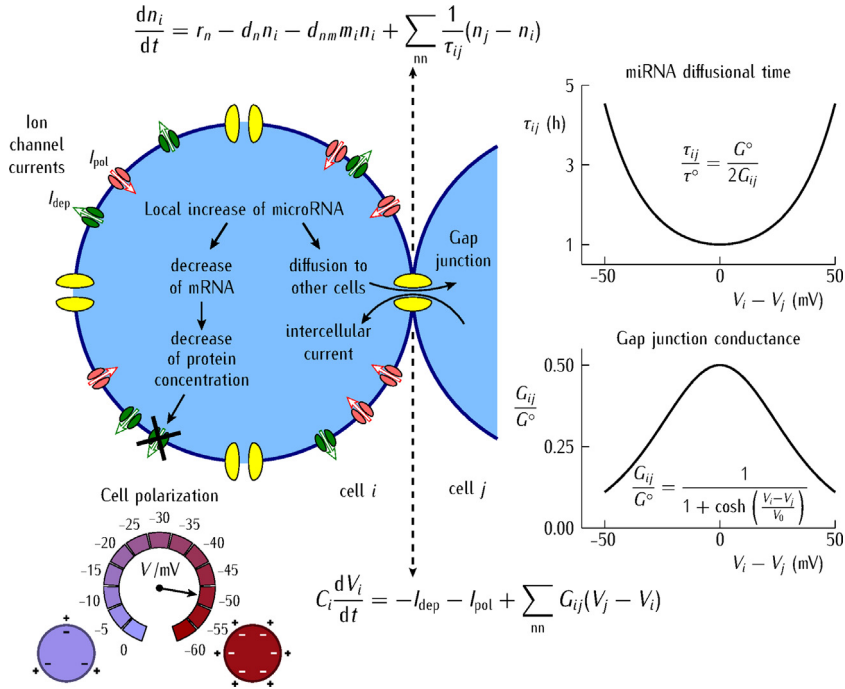


Fig. 23. Extension of the model of Fig. 4 for the coupling of the central cell i to the neighboring cell j through intercellular voltage-gated junctions of dimensionless conductance G_{ij}/G^o . The experimental bell-shaped conductance is obtained with the maximum conductance $G^o/G_{pol}^o = 1$ and the width potential $V_0 = 18$ mV. We assume that the miRNA diffusional time decreases with the junction conductance as $\tau_{ij}/\tau^o = G^o/2G_{ij}$, where $\tau^o = 1$ h is a reference time. The coupling equations for n_i and V_i extend the single-cell interplay between the transcriptional and bioelectrical signals to the multicellular level case.

Source: Reprinted figure from Ref. [44].

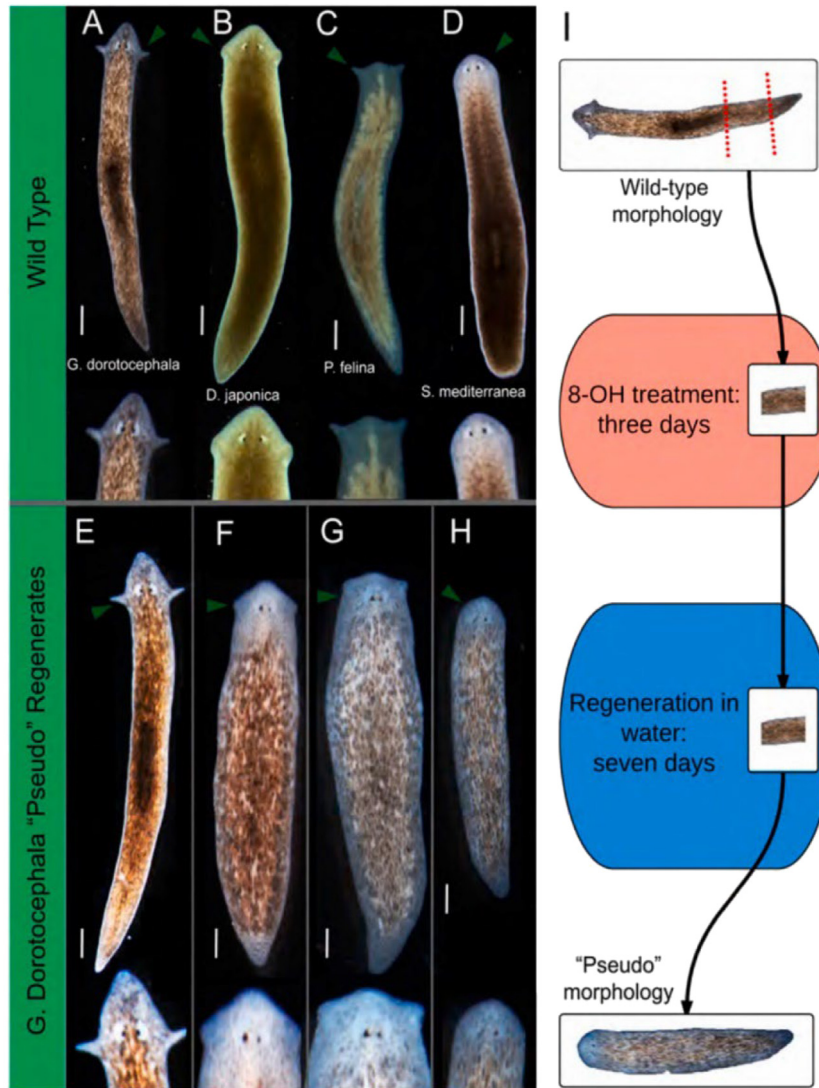


Fig. 24. The distinct head morphologies that are observed in four planaria flatworm species after octanol treatment (A–D). The arrows show the different head shapes (E–H) obtained for the pre-tail fragments that are amputated from *G. dorotocephala* worms and treated following the junction blocking procedure shown. The scale bar corresponds to 0.5 mm.

Source: Reprinted figure from Ref. [30].

Experimentally, when the junctions are blocked by the addition of an external molecule, the local concentrations of the morphogens diffusing through them are changed, which leads to modifications in the instructive patterns that modulate morphological outcomes such as the head shape in regenerating flatworms [30,85]. Figs. 24 (experiments) and 25 (model simulations) suggest that the modulation of the intercellular communication can produce different anatomies from the same genome without changing complex gene-regulatory networks. In this way, the reprogramming of spatial differentiation and biological morphologies is feasible by the external manipulation of the gap junctions and the resulting changes in the spatial distribution of instructive biochemical signals [30,31,85].

We analyze now the electrical interaction between cell membrane potentials and charged nanoparticles [22,109,110]. In clinical applications, the distribution of nanoparticles over small tissue regions can be modulated not only by local hydrophobic effects but also by the spatial pattern of electric potentials. In the case of charged surfaces, the binding of the nanoparticles to the membrane could modify the individual cell potentials. Flux cytometry, dyes, and fluorescence microscopy techniques show indeed that positively charged amine-modified polystyrene nanoparticles from the external microenvironment are bound to the membrane of Chinese Hamster Ovary (CHO) cells causing significant depolarization [109]. Bioelectrical models can also provide qualitative physical insights in this case. Fig. 26 shows that the number of positive nanoparticles around each individual cell scales to the negative electrical potentials [22], thus

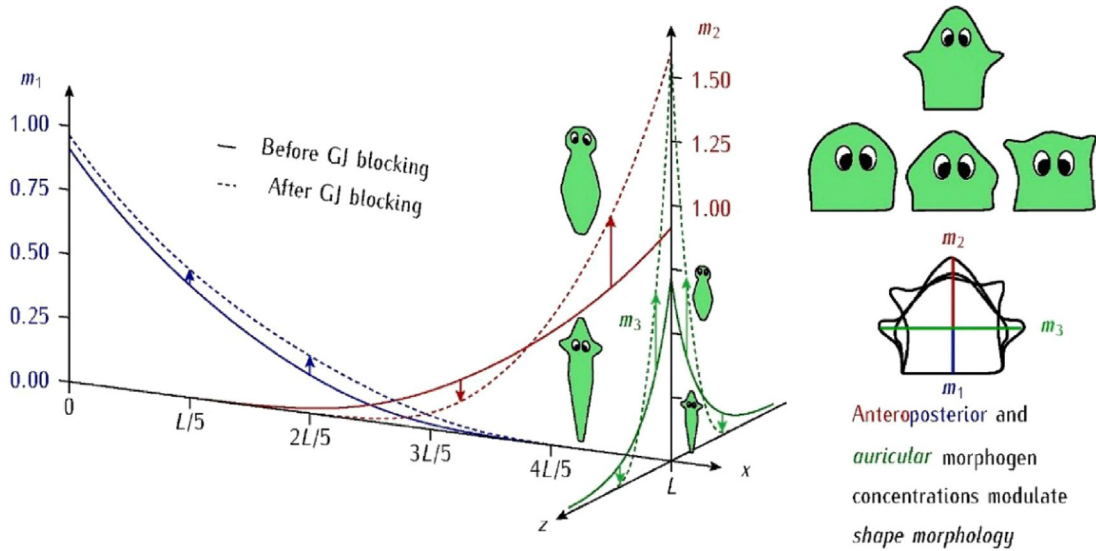


Fig. 25. When diffusion coefficients of the anteroposterior morphogens 1 and 2 regulating the flatworm shape decrease because of the blocking of the intercellular gap junctions (GJ), the spatial patterns of the respective morphogen concentrations are modified (arrows), as schematically shown. These spatial changes, together with that of the auricular morphogen diffusion, modify the original instructive patterns, which leads to the distinct head shapes (right) of Fig. 24 [30]. The extended model equations have been described previously [85].
Source: Reprinted figure from Ref. [85].

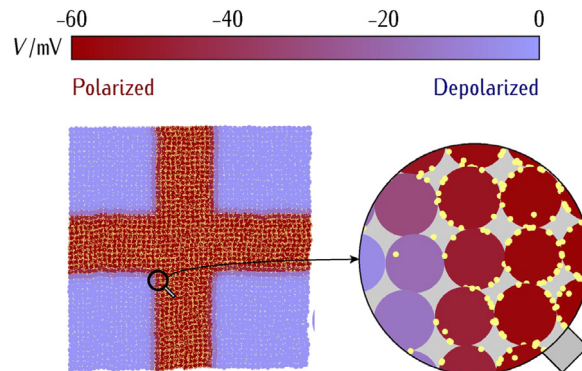


Fig. 26. The simulation suggests that the spatial distribution of positively charged nanoparticles (small yellow circles) closely follows the cell potential patterns: particle accumulation is more significant around polarized (more negative) cells than around depolarized cells [22]. For clarity, the inset zooms a small region of the multicellular system. Experimentally, the introduction of these nanoparticles can be used to externally modulate cell polarization [109,110].
Source: Reprinted figure from Ref. [22].

suggesting local disruptions of the potential difference between the membrane inside and outside. Experimentally, these phenomena may provide methods based on charged nanoprobe to identify and modulate electric potential patterns. A limitation here is that the surface properties of the nanoparticles in the vicinity of the cell membrane can be severely altered, affecting cellular uptake and sensing, because protein coronas develop on the nanoparticle surfaces [110].

To show how bioelectrical and biomechanical effects can be coupled together [78], we discuss briefly now a possible connection between lipid organization and cell proliferation [111]. Fig. 27 suggests that the concerted action of the *pol* and *dep* ion channel families inserted in the cell membrane can be described as a biological field-effect transistor regulated by the cell potential [111]. Remarkably, it could be the bioelectrical cell depolarization that triggers the downstream biomechanical and biochemical processes controlling mitogenic signals via specific lipids and proteins (Fig. 27). This fact shows again the master role of bioelectrical signals that we have tried to emphasize in this report.

7. Discussion, open problems, and perspectives

How can groups of cells cooperate in anatomical structures? How can collective decision-making be established at multiple scales? Cell and multicellular states can be associated with instructive memories across multiple scales, e.g. mathematical attractors in single-cell genetic regulatory networks [90,91], morphogen gradients and positional information

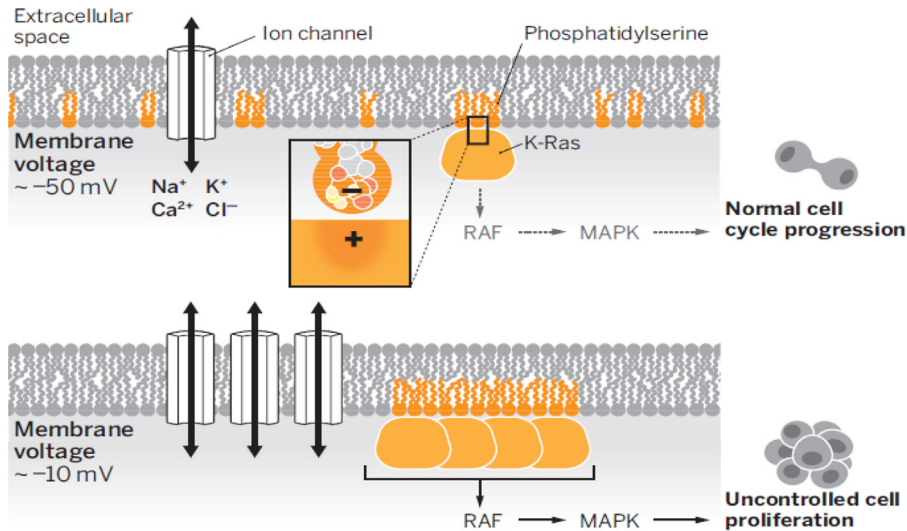


Fig. 27. The cell potential (membrane voltage here) can modify the lipid organization, with downstream consequences on the biochemical networks regulating mitogenic signaling circuits. Nonproliferating cells (top) tend to be in the polarized state modulated by the membrane ion channels. The negatively charged phosphatidylserine lipids form small clusters that localize the positively charged residues of the signaling protein K-Ras. This fact results in a low activation of the RAF-MAPK biochemical path [111,112]. However, a depolarizing channel over-expression acts to depolarize the cell (bottom), thus increasing the lipids and protein K-Ras clustering, which promotes uncontrolled proliferation.

Source: Reprinted figure from A. Acardi, Lipids link ion channels and cancer, Science 349 (2015) 789–790, with permission from AAAS.

in tissues [5,113], and target anatomies in model animals [9]. In addition, the spatio-temporal patterns of electrical current [36,39] and potential [8,9] are also instructive for development and regeneration (Fig. 1). The model simulations described here emphasize the interplay between transcriptional and bioelectrical patterns, together with the role of the intercellular coupling heterogeneities, in the establishment of instructive modules at the system-level. At the single-cell level, a particular complement of ion channel proteins determines the bioelectrical state as a local memory. Detailed technical descriptions of the numerical solution and software used in the simulations can be found in [15,22,44]. Alternatively, a complete simulation of the bioelectrical dynamics has been implemented by Pietak and Levin using the BETSE simulator [82].

We have focused our study on an important but limited number of channels and chosen not to address system-specific biochemical and biomechanical mechanisms that could be associated with the memory patterns [78,82]. Thus, we could not claim a quantitative description of the myriad of spatio-temporal cascades and indirect effects characteristic of developmental bioelectricity, which may include also non-conductive channel protein roles [35]. Experimentally, however, the fact is that potassium, sodium, and calcium ion channels, together with intercellular junctions such as connexins, are ubiquitous in cell systems. Also, these transporters are functionally interrelated [10] and tend to show voltage-gated conductances because of both transcriptional and post-translational mechanisms (Figs. 7 and 12). In addition, it is the difference between the electric potential of the different modules across the multicellular system, not the specific channels and junctions involved, that is relevant for the memory patterns orchestrating development (Fig. 1), as shown by the ability of specific bioelectric perturbations to induce ectopic eyes, tail regeneration, and brain repair (9). In particular, calcium levels, which are crucial to cellular communication, are intrinsically coupled with cell potentials in different ion transporters context [34–38,48,114]. Also, the observation that changes in the intercellular connectivity can lead to transitions between multicellular states has a solid experimental basis [30,31,33,69,92,93,114] and emphasizes bioelectrical integration-segregation as a causal multicellular pattern that leads to morphological effects [8,9].

In view of this clear evidence, our *bioelectrical metaphor* is based on two simple ideas: (i) cells expressing typical ion channels can keep stable memories based on polarized/depolarized single-cell potentials (Figs. 2, 5, and 6) and (ii) the intercellular junctions can couple together these individual cell states to give *labile but stable* multicellular patterns (Figs. 12–16, and 17b). These junctions provide a *short range integration* within regions of similar cell potentials and a *long range segregation* between subsystems of different cell potentials. The *bioelectrical modules*, together with the particular *integration-segregation topology* between them, defines the multicellular pattern memory (Fig. 1) that can be retrieved or rewritten because of the intercellular electrical coupling. In this context, it is important to realize that the individual bioelectrical modules as a whole do not need to know the global system outcome to be eventually established: it is the integration-segregation topology established under the appropriate external signals and internal correlations that will lead to the developmental compartments characteristic of a particular morphology.

Biological systems are open to input signals from the external microenvironment. The different developmental phases and regeneration stages show distinct spatial distributions of potassium and calcium ions, neurotransmitters, and other

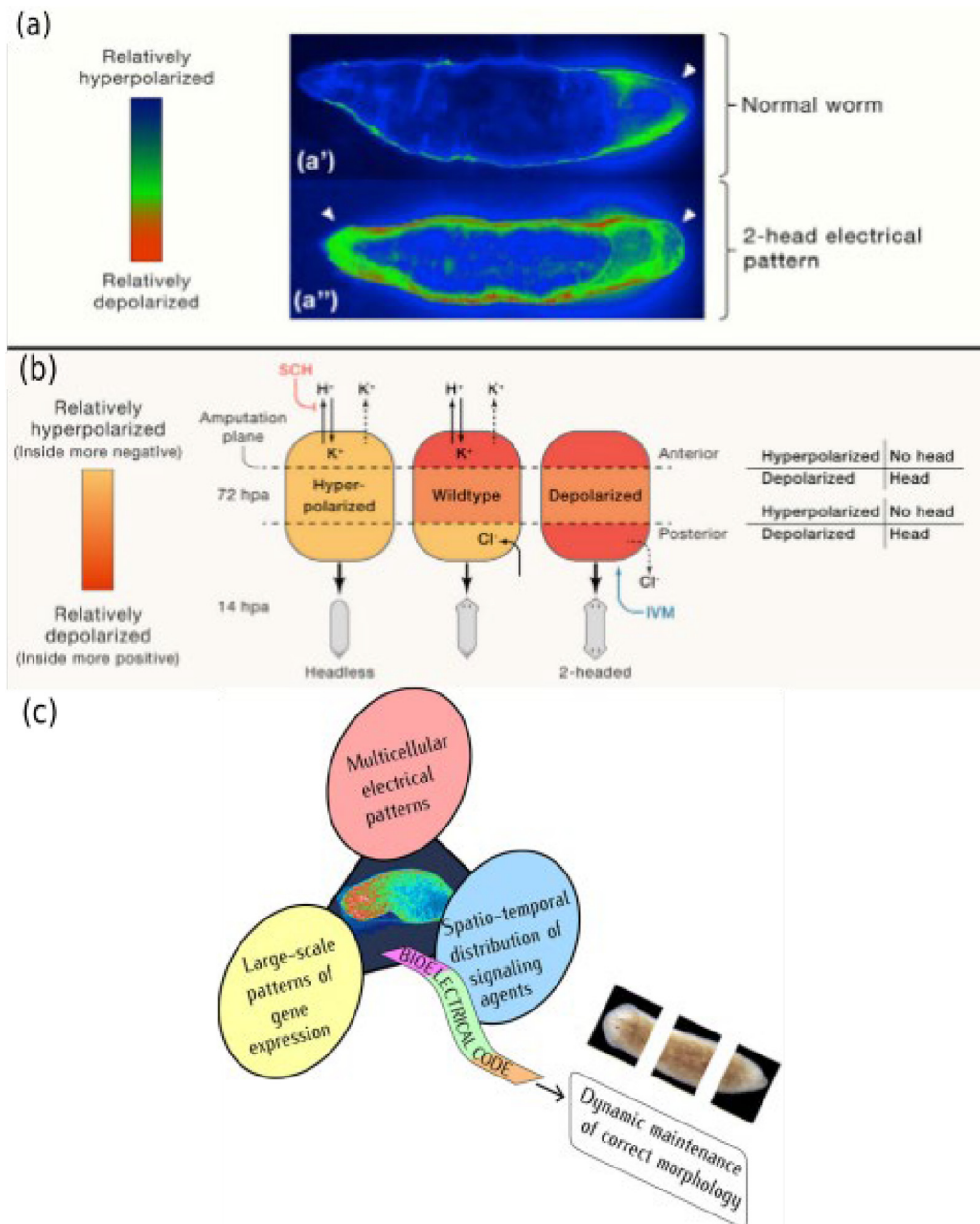


Fig. 28. Bioelectrical patterns and dynamic maintenance of morphology. (a) Bioelectrical patterns of cell potentials showing a depolarized region in wild-type worms (a') and a bipolar pattern in worms that are going two-headed (a'') [9]. (b) In this case, the bioelectrical circuit composed of potassium and chloride channels, together with a proton/potassium exchanger pump and its inhibitor (SCH), is instructive for the worm anatomy. The use of RNAi or channel drugs can modulate the cell polarization patterns, alter the downstream expression of axial polarity genes, and produce two-headed or no-headed worms [9]. (c) Thus, a *bioelectrical code* [20] results from the correlation between the multicellular electrical patterns and the spatio-temporal distribution of signaling ions and molecules that modulate large-scale expression patterns [8,9]. This biophysical-biochemical feedback is central to regeneration and dynamic maintenance of correct morphology.
Source: Reprinted figures from Refs. [8,9].

transcription activators that are correlated with the electric potential patterns (Fig. 28), thus establishing a *bioelectrical code* [20]. It is this correlation that shapes the bioelectrical pattern memories of Fig. 1 where electrical signals act as templates for slow biochemical signals following a hierarchy of experimental times [115]. Because cell potentials can influence protein expression (Figs. 3, 7, and 10), actions based on the pharmacological manipulation of the channels conductances and ionic gradients constitute experimentally plausible methods to modify the *bioelectrical memory pattern*

encoded by these potentials. Some external actions are at the *hardware* level, e.g., the microinjection of a mRNA corresponding to a specific channel protein [15], while other are at the *software* level, e.g., the introduction of a molecule causing a transient intercellular junction blocking [30] or the use of optogenetics to provide a temporary bioelectric pattern stimulus [116,117]. Clearly, these actions should be local and are not without clinical risks because channels regulate multiple functions (e.g. the cardiac rhythm) and are involved in multitude of cell processes (e.g. adhesion, volume regulation, and apoptosis).

Certainly, because of the coupling of bioelectricity with additional biochemical and biomechanical mechanisms [118–127], more experimental research on causation concerning specific physical mechanisms that modulate collective gene expression patterns is needed [9,10,120]. By knowing these mechanisms, the manipulation of endogenous electric potentials at the multicellular level could assist the exceedingly difficult problem of managing those intracellular and intercellular gene regulatory mechanisms that depend on cell potentials. In this context, the simulations discussed here have clear practical predictions: by identifying different bioelectrical patterns, target memories can be retrieved or externally rewritten by means of appropriate experimental actions on the individual modules and their integration-segregation topology [15,19,22,76]. Morphological consequences immediately follow because *modularity* and *integration* are observed at multiple scales [9,10,17,121–124], being crucial to biological information processing. Similarly, neural functions emerge only at the ensemble level where individual cells are integrated into plastic multicellular patterns that go beyond single neuron states [125].

We believe that the bioelectric model described offers useful qualitative insights into the experimental interplay between transcription and bioelectricity. In particular, the memory patterns suggest new opportunities at the *intermediate scale* characteristic of endogenous multicellular fields, which may constitute a complementary approach to the *low scale* case of manipulating cell molecular mechanisms and the *large scale* case of applying electrical fields to tissues and organs.

CRediT authorship contribution statement

Javier Cervera: Methodology, Software, Formal analysis, Writing – review & editing. **Michael Levin:** Conceptualization, Formal analysis, Writing – review & editing. **Salvador Mafe:** Conceptualization, Methodology, Formal analysis, Writing – original draft.

Declaration of competing interest

M.L. is cofounder of the company Morphoceuticals Inc., which operates in the regenerative medicine and hopes to do regeneration by bioelectric technology. The authors declare no other competing interests.

Acknowledgments

M.L. acknowledges the support by the Templeton World Charity Foundation (TWCF0606) and the Guy Foundation Family Trust 103733-00001.

References

- [1] B. Hille, *Ion Channels of Excitable Membranes*, Sinauer Associates, Sunderland, 1992.
- [2] R.A. Gatenby, The role of cell membrane information reception, processing, and communication in the structure and function of multicellular tissue, *Int. J. Mol. Sci.* 20 (2019) 3609.
- [3] K. Robinson, M. Messerli, Electric embryos: the embryonic epithelium as a generator of developmental information, in: C. McCaig (Ed.), *Nerve Growth and Guidance*, Portland Press, Portland, 1996, pp. 131–141.
- [4] Y. Dang, D.A.J. Grundel, H. Youk, Cellular dialogues: Cell-cell communication through diffusible molecules yields dynamic spatial patterns, *Cell Syst.* 10 (2020) 82–98.
- [5] H. Meinhardt, A. Gierer, Pattern formation by local self-activation and lateral inhibition, *BioEssays* 22 (2000) 753–760.
- [6] P. Müller, E. El-Sherif, A systems-level view of pattern formation mechanisms in development, *Dev. Biol.* 460 (2020) 1.
- [7] K.A. McLaughlin, M. Levin, Bioelectric signaling in regeneration: Mechanisms of ionic controls of growth and form, *Dev. Biol.* 433 (2018) 177–189.
- [8] J. Cervera, M. Levin, S. Mafe, Bioelectrical coupling of single-cell states in multicellular systems, *J. Phys. Chem. Lett.* 11 (2020) 3234–3241.
- [9] M. Levin, Bioelectric signaling: Reprogrammable circuits underlying embryogenesis, regeneration, and cancer, *Cell* 184 (2021) 1971.
- [10] M.P. Harris, Bioelectric signaling as a unique regulator of development and regeneration, *Development* 148 (2021) dev180794.
- [11] J.R. Lazzari-Dean, A.M.M. Gest, E.W. Miller, Measuring absolute membrane potential across space and time, *Annu. Rev. Biophys.* 50 (2021) 447–468.
- [12] F. Durant, J. Bischof, C. Fields, J. Morokuma, J. LaPalme, A. Hoi, M. Levin, The role of early bioelectric signals in the regeneration of planarian anterior/posterior polarity, *Biophys. J.* 116 (2019) 948.
- [13] L.N. Vandenberg, R.D. Morrie, D.S. Adams, V-ATPase-dependent ectodermal voltage and pH regionalization are required for craniofacial morphogenesis, *Dev. Dyn.* 240 (2011) 1889–1904.
- [14] B.T. Chernet, M. Levin, Transmembrane voltage potential is an essential cellular parameter for the detection and control of tumor development in a *Xenopus* model, *Dis. Model. Mech.* 6 (2013) 595–607.
- [15] V.P. Pai, J. Cervera, S. Mafe, V. Willocq, E.K. Lederer, M. Levin, HCN2 channel-induced rescue of brain teratogenesis via local and long-range bioelectric repair, *Front. Cell. Neurosci.* 14 (2020) 136.
- [16] M. Levin, Molecular bioelectricity: how endogenous voltage potentials control cell behavior and instruct pattern regulation in vivo, *Mol. Biol. Cell* 25 (2014) 3835–3850.

- [17] S.F. Gilbert, J.M. Opitz, R.A. Raff, Resynthesizing evolutionary and developmental biology, *Dev. Biol.* 173 (1996) 357–372.
- [18] M. Levin, Morphogenetic fields in embryogenesis, regeneration, and cancer: non-local control of complex patterning, *Biosystems* 109 (2012) 243–261.
- [19] J. Cervera, V.P. Pai, M. Levin, S. Mafe, From non-excitable single-cell to multicellular bioelectrical states supported by ion channels and gap junction proteins: Electrical potentials as distributed controllers, *Prog. Biophys. Mol. Biol.* 149 (2019) 39.
- [20] M. Levin, C.J. Martyniuk, The bioelectric code: An ancient computational medium for dynamic control of growth and form, *Biosystems* 164 (2018) 76–93.
- [21] S.L. Payne, P. Ram, D.H. Srinivasan, T.T. Le, M. Levin, M.J. Oudin, Potassium channel-driven bioelectric signalling regulates metastasis in triple-negative breast cancer, *EBioMedicine* 75 (2022) 103767.
- [22] J. Cervera, A. Alcaraz, S. Mafe, Bioelectrical signals and ion channels in the modeling of multicellular patterns and cancer biophysics, *Sci. Rep.* (2016) 20403.
- [23] J. Cervera, P. Ramirez, M. Levin, S. Mafe, Community effects allow bioelectrical reprogramming of cell membrane potentials in multicellular aggregates: Model simulations, *Phys. Rev. E* 102 (2020) 052412.
- [24] A. Rioli, J. Cervera, M. Levin, S. Mafe, Cell systems bioelectricity: How different intercellular gap Junctions Could Regionalize a multicellular aggregate, *Cancers (Basel)* 13 (2021) 5300.
- [25] J. Cervera, J.A. Manzanares, S. Mafe, M. Levin, Synchronization of bioelectric oscillations in networks of nonexcitable cells: From single-cell to multicellular states, *J. Phys. Chem. B* 123 (2019) 3924–3934.
- [26] M.B. Bhavsar, L. Leppik, K.M. Costa Oliveira, J.H. Barker, Role of bioelectricity during cell proliferation in different cell types, *Front. Bioeng. Biotechnol.* 8 (2020) 603.
- [27] R.A. Gatenby, B.R. Frieden, Cellular information dynamics through transmembrane flow of Ions, *Sci. Rep.* 7 (2017) 15075.
- [28] J. Cervera, A. Alcaraz, S. Mafe, Membrane potential bi-stability in non-excitable cells as described by inward and outward voltage-gated ion channels, *J. Phys. Chem. B* 11 (2014) 12444–12450.
- [29] R. Law, M. Levin, Bioelectric memory: modeling resting potential bistability in amphibian embryos and mammalian cells, *Theor. Biol. Med. Model.* 12 (2015) 22.
- [30] M. Emmons-Bell, F. Durant, J. Hammelman, N. Bessonov, V. Volpert, J. Morokuma, K. Pinet, D.S. Adams, A. Pietak, D. Lobo, M. Levin, Gap junctional blockade stochastically induces different species-specific head anatomies in genetically wild-type *Girardia dorotocephala* flatworms, *Int. J. Mol. Sci.* 16 (2015) 27865.
- [31] C.M. Glen, T.C. McDevitt, M.L. Kemp, Dynamic intercellular transport modulates the spatial patterning of differentiation during early neural commitment, *Nature Commun.* 9 (2018) 4111.
- [32] J. Cervera, J.A. Manzanares, S. Mafe, Electrical coupling in ensembles of nonexcitable cells: Modeling the spatial map of single cell potentials, *J. Phys. Chem. B* 119 (2015) 2968–2978.
- [33] M. Levin, G. Pezzulo, J.M. Finkelstein, Endogenous bioelectric signaling networks: Exploiting voltage gradients for control of growth and form, *Annu. Rev. Biomed. Eng.* 19 (2017) 353–387.
- [34] O.V. Aslanidi, O.A. Mornev, O. Skyggebjerg, P. Arkhammar, O. Thastrup, M.P. Sørensen, P.L. Christiansen, K. Conradsen, A.C. Scott, Excitation wave propagation as a possible mechanism for signal transmission in pancreatic islets of Langerhans, *Biophys. J.* 80 (2001) 1195–1209.
- [35] L.F. George, E. Bates, Mechanisms underlying influence of bioelectricity in development, *Front. Cell Dev. Biol.* 10 (2022) 772230.
- [36] B. Reid, M. Zhao, The electrical response to injury: Molecular mechanisms and wound healing, *Adv. Wound Care (New Rochelle)* 3 (2014) 184–201.
- [37] T. O’Leary, A.H. Williams, A. Franci, E. Marder, Cell types, network homeostasis, and pathological compensation from a biologically plausible ion channel expression model, *Neuron* 82 (2014) 809–821.
- [38] L. Catacuzzeno, B. Fioretti, F. Franciolini, A theoretical study on the role of Ca(2+)-activated K⁺ channels in the regulation of hormone-induced Ca²⁺ oscillations and their synchronization in adjacent cells, *J. Theoret. Biol.* 309 (2012) 103–112.
- [39] L.F. Jaffe, The role of ionic currents in establishing developmental pattern, *Philos. Trans. R. Soc. B* 295 (1981) 553–566.
- [40] D. Capek, P. Müller, Positional information and tissue scaling during development and regeneration, *Development* 146 (2019) dev177709.
- [41] S.Y. Shvartsman, R.E. Baker, Mathematical models of morphogen gradients and their effect on gene expression, *WIREs Dev. Biol.* 1 (2012) 715–730.
- [42] C. Williamson, H.M. Chamberlin, A.T. Dawes, Coordination of local and long range signaling modulates developmental patterning, *J. Theoret. Biol.* 517 (2021) 110596.
- [43] X. Wang, M.L. Veruki, N.V. Bukoreshtliev, E. Hartveit, H.H. Gerdes, Animal cells connected by nanotubes can be electrically coupled through interposed gap-junction channels, *Proc. Natl. Acad. Sci. USA* 107 (2010) 17194–17199.
- [44] J. Cervera, S. Meseguer, S. Mafe, MicroRNA intercellular transfer and bioelectrical regulation of model multicellular ensembles by the gap junction connectivity, *J. Phys. Chem. B* 121 (2017) 7602.
- [45] V.P. Pai, J.M. Lemire, J.F. Pare, G. Lin, Y. Chen, M. Levin, Endogenous gradients of resting potential instructively pattern embryonic neural tissue via Notch signaling and regulation of proliferation, *J. Neurosci.* 35 (2015) 4366–4385.
- [46] F. Durant, J. Morokuma, C. Fields, K. Williams, D.S. Adams, M. Levin, Long-term, stochastic editing of regenerative anatomy via targeting endogenous bioelectric gradients, *Biophys. J.* 112 (2017) 2231–2243.
- [47] J. Cervera, S. Meseguer, M. Levin, S. Mafe, Bioelectrical model of head-tail patterning based on cell ion channels and intercellular gap junctions, *Bioelectrochemistry* 132 (2020) 107410.
- [48] P. McMillen, R. Novak, M. Levin, Toward decoding bioelectric events in xenopus embryogenesis: New methodology for tracking interplay between calcium and resting potentials in vivo, *J. Mol. Biol.* 432 (2020) 605–620.
- [49] V.P. Pai, S. Aw, T. Shomrat, J.M. Lemire, M. Levin, Transmembrane voltage potential controls embryonic eye patterning in xenopus laevis, *Development* 139 (2012) 313–323.
- [50] D. Zuo, K. Chen, M. Zhou, Z. Liu, H. Chen, Kir2.1 and K2P1 channels reconstitute two levels of resting membrane potential in cardiomyocytes, *J. Physiol.* 595 (2017) 5129–5142.
- [51] S.R. Williams, S.R. Christensen, G.J. Stuart, M. Häusser, Membrane potential bistability is controlled by the hyperpolarization-activated current I(H) in rat cerebellar Purkinje neurons in vitro, *J. Physiol.* 539 (2002) 469–483.
- [52] H. van Mil, J. van Heukelom, M. Bier, A bistable membrane potential at low extracellular potassium concentration, *Biophys. Chem.* 106 (2003) 15–21.
- [53] A. Moshkforoush, B. Ashenagar, O.F. Harraz, F. Dabertrand, T. Longden, M.T. Nelson, N.M. Tsoukias, The capillary Kir channel as sensor and amplifier of neuronal signals: Modeling insights on K⁺-mediated neurovascular communication, *Proc. Natl. Acad. Sci. USA* 117 (2020) 16626–16637.
- [54] J. Golowasch, A. Bose, Y. Guan, D. Salloum, A. Roeser, F. Nadim, A balance of outward and linear inward ionic currents is required for generation of slow-wave oscillations, *J. Neurophysiol.* 118 (2017) 1092–1104.
- [55] R.D. Kirkton, N. Bursac, Engineering biosynthetic excitable tissues from unexcitable cells for electrophysiological and cell therapy studies, *Nature Commun.* 2 (2011) 300.

- [56] H.M. McNamara, R. Salegame, Z.A. Tanoury, H. Xu, S. Begum, G. Ortiz, O. Pourquie, A.E. Cohen, Bioelectrical domain walls in homogeneous tissues, *Nat. Phys.* 16 (2020) 357.
- [57] R.C. Stan, D.K. Bhatt, M.M. de Camargo, Cellular adaptation relies on regulatory proteins having episodic memory, *Bioessays* 42 (2020) 1900115.
- [58] A. Koseska, P.I.H. Bastiaens, Cell signaling as a cognitive process, *EMBO J.* 36 (2017) 568–582.
- [59] J. Cervera, S. Meseguer, S. Mafe, The interplay between genetic and bioelectrical signaling permits a spatial regionalisation of membrane potentials in model multicellular ensembles, *Sci. Rep.* 6 (2016) 35201.
- [60] J. Golowasch, Neuronal homeostasis: Voltage brings it all together, *Curr. Biol.* 29 (2019) R641–R646.
- [61] V.R. Rao, M. Perez-Neut, S. Kaja, S. Gentile, Voltage-gated ion channels in cancer cell proliferation, *Cancers (Basel)* 7 (2015) 849.
- [62] M. Yang, W.J. Brackenbury, Membrane potential and cancer progression, *Front. Physiol.* 4 (2013) 185.
- [63] D.J. Blackiston, K.A. McLaughlin, M. Levin, Bioelectric controls of cell proliferation: Ion channels, membrane voltage and the cell cycle, *Cell Cycle* 8 (2009) 3527.
- [64] D. Urrego, A.P. Tomczak, F. Zahed, W. Stühmer, L.A. Pardo, Potassium channels in cell cycle and cell proliferation, *Philos. Trans. R. Soc. B* 369 (2014) 20130094.
- [65] P.G. Barghouth, M. Thiruvalluvan, N.J. Oviedo, Bioelectrical regulation of cell cycle and the planarian model system, *Biochim. Biophys. Acta* 1848 (2015) 2629–2637.
- [66] C. Giorgi, A. Danese, S. Missiroli, S. Patergnani, P. Pinton, Calcium dynamics as a machine for decoding signals, *Trends Cell Biol.* 28 (2018) 258–273.
- [67] K. Krawczyk, S. Xue, P. Buchmann, G. Charpin-El-Hamri, P. Saxena, M.D. Hussherr, J. Shao, H. Ye, M. Xie, M. Fussenegger, Electrogenetic cellular insulin release for real-time glycemic control in type 1 diabetic mice, *Science* 368 (2020) 993–1001.
- [68] L. Adermark, D.M. Lovinger, Electrophysiological properties and gap junction coupling of striatal astrocytes, *Neurochem. Int.* 52 (2008) 1365–1372.
- [69] B. Ma, R. Buckalew, Y. Du, C.M. Kiyoshi, C.C. Alford, W. Wang, D.M. McTigue, J.J. Enyeart, D. Terman, M. Zhou, Gap junction coupling confers isopotentiality on astrocyte syncytium, *Glia* 64 (2016) 214–226.
- [70] T. Martins-Marques, T. Ribeiro-Rodrigues, D. Batista-Almeida, T. Aasen, B.R. Kwak, H. Girao, Biological functions of connexin43 beyond intercellular communication, *Trends Cell Biol.* 29 (2019) 835–847.
- [71] T.M. Ribeiro-Rodrigues, T. Martins-Marques, S. Morel, B.R. Kwak, H. Girão, Role of connexin 43 in different forms of intercellular communication - gap junctions, extracellular vesicles and tunnelling nanotubes, *J. Cell Sci.* 130 (2017) 3619–3630.
- [72] N. Palacios-Prado, F.F. Bukauskas, Heterotypic gap junction channels as voltage-sensitive valves for intercellular signaling, *Proc. Natl. Acad. Sci. USA* 106 (2009) 14855–14860.
- [73] T. Desplantez, D. Halliday, E. Dupont, R. Weingart, Cardiac connexins Cx43 and Cx45: formation of diverse gap junction channels with diverse electrical properties, *Pflügers Arch.* 448 (2004) 363–375.
- [74] X. Lin, Q. Xu, R.D. Veenstra, Functional formation of heterotypic gap junction channels by connexins-40 and -43, *Channels* 8 (2014) 433–443.
- [75] M. Hitomi, L.P. Deleyrolle, E.E. Mulkearns-Hubert, A. Jarrar, M. Li, M. Sinyuk, B. Otvos, S. Brunet, W.A. Flavahan, C.G. Hubert, W. Goan, J.S. Hale, A.G. Alvarado, A. Zhang, M. Rohaus, M. Oli, V. Vedam-Mai, J.M. Fortin, H.S. Futch, B. Griffith, Q. Wu, C.H. Xia, X. Gong, M.S. Ahluwalia, J.N. Rich, B.A. Reynolds, J.D. Lathia, Differential connexin function enhances self-renewal in glioblastoma, *Cell Rep.* 11 (2015) 1031–1042.
- [76] J. Cervera, S. Meseguer, S. Mafe, Intercellular connectivity and multicellular bioelectric oscillations in nonexcitable cells: A biophysical model, *ACS Omega* 3 (2018) 13567–13575.
- [77] V.P. Pai, A. Pietak, V. Willocq, B. Ye, N.-Q. Shi, M. Levin, HCN2 rescues brain defects by enforcing endogenous voltage pre-patterns, *Nature Commun.* 9 (2018) 998.
- [78] A. Leronna, L. Bardella, L. Dorfmann, A. Pietak, M. Levin, On the coupling of mechanics with bioelectricity and its role in morphogenesis, *J. R. Soc. Interface* 17 (2020) 0177.
- [79] H.T. Vu, S. Mansour, M. Kücken, C. Blasse, C. Basquin, J. Azimzadeh, E.W. Myers, L. Brusch, J.C. Rink, Dynamic polarization of the multiciliated planarian epidermis between body plan landmarks, *Dev. Cell.* 51 (2019) 526–542.
- [80] S. Herath, D. Lobo, Cross-inhibition of turing patterns explains the self-organized regulatory mechanism of planarian fission, *J. Theoret. Biol.* 485 (2020) 110042.
- [81] A. Pietak, J. Bischof, J. LaPalme, J. Morokuma, M. Levin, Neural control of body-plan axis in regenerating planaria, *PLoS Comput. Biol.* 15 (2019) e1006904.
- [82] A. Pietak, M. Levin, Bioelectric gene and reaction networks: computational modelling of genetic, biochemical and bioelectrical dynamics in pattern regulation, *J. R. Soc. Interface* 14 (2017) 20170425.
- [83] J. Cervera, A. Pietak, M. Levin, S. Mafe, Bioelectrical coupling in multicellular domains regulated by gap junctions: A conceptual approach, *Bioelectrochem.* 123 (2018) 45–61.
- [84] T. Adell, F. Cebria, E. Salo, Gradients in planarian regeneration and homeostasis, *Cold Spring Harb. Perspect. Biol.* 2 (2010) a000505.
- [85] J. Cervera, M. Levin, S. Mafe, Morphology changes induced by intercellular gap junction blocking: A reaction–diffusion mechanism, *BioSystems* 209 (2021) 104511.
- [86] A. Prindle, J. Liu, M. Asally, S. Ly, J. Garcia-Ojalvo, G.M. Süel, Ion channels enable electrical communication in bacterial communities, *Nature* 527 (2015) 59–63.
- [87] C.-Y. Yang, M. Bialecka-Fornal, C. Weatherwax, J.W. Larkin, A. Prindle, J. Liu, J. Garcia-Ojalvo, G.M. Süel, Encoding membrane-potential-based memory within a microbial community, *Cell Syst.* 10 (2017) 417–423.
- [88] A. Stanoev, C. Schröter, A. Koseska, Robustness and timing of cellular differentiation through population-based symmetry breaking, *Development* 148 (2021) dev197608.
- [89] V.P. Zhdanov, Kinetic models of gene expression including non-coding RNAs, *Phys. Rep.* 500 (2011) 1–42.
- [90] C.H. Waddington, *The Strategy of the Genes. A Discussion of Some Aspects of Theoretical Biology*, Allen & Unwin, London, 1957.
- [91] S. Huang, The molecular and mathematical basis of waddington's epigenetic landscape: A framework for post-darwinian biology? *BioEssays* 34 (2012) 149–157.
- [92] D. Banerjee, Connexin's connection in breast cancer growth and progression, *Int. J. Cell Biol.* 2016 (2016) 9025905.
- [93] M. Mesnil, T. Aasen, J. Boucher, A. Chepied, L. Cronier, N. Defamie, P. Kameritsch, D.W. Laird, P.D. Lampe, J.D. Lathia, E. Leithe, P.P. Mehta, A. Monvoisin, K. Pogoda, W.-C. Sin, A. Taberner, H. Yamasaki, E.S. Yeh, M.L.Z. Dagli, C.C. Naus, An update on minding the gap in cancer, *Biochim. Biophys. Acta* 1860 (2018) 237–243.
- [94] B.T. Chernet, C. Fields, M. Levin, Long-range gap junctional signaling controls oncogene-mediated tumorigenesis in xenopus laevis embryos, *Front. Physiol.* 5 (2015) 519.
- [95] A. Goldbeter, Biological rhythms as temporal dissipative structures, in: S.A. Rice (Ed.), *Special Volume in Memory of Ilya Prigogine*, in: *Adv. Chem. Phys.*, vol. 135, Wiley, 2007, pp. 253–293.
- [96] O. Decroly, A. Goldbeter, Birhythmicity, chaos, and other patterns of temporal self-organization in a multiply regulated biochemical system, *Proc. Natl. Acad. Sci. USA* 79 (1982) 6917–6921.
- [97] J. Yan, A. Goldbeter, Multi-rhythmicity generated by coupling two cellular rhythms, *J. R. Soc. Interface* 16 (2019) 20180835.

- [98] P.R.F. Rocha, P. Schlett, L. Schneider, M. Dröge, V. Mailänder, H.L. Gomes, P.W.M. Blom, D.M. de Leeuw, Low frequency electric current noise in glioma cell populations, *J. Mater. Chem. B* 3 (2015) 5035–5039.
- [99] M. Ribeiro, A. Elghajji, S.P. Fraser, Z.D. Burke, D. Tosh, M.B.A. Djamgoz, P.R.F. Rocha, Human breast cancer cells demonstrate electrical excitability, *Front. Neurosci.* 14 (2020) 404.
- [100] T.H. Hraha, M.J. Westacott, M. Pozzoli, A.M. Notary, P.M. McClatchey, R.K.P. Benninger, Phase transitions in the multicellular regulatory behavior of pancreatic islet excitability, *PLoS Comput. Biol.* 10 (2014) e1003819.
- [101] C.D. Tsiariris, A. Aulehla, Self-organization of embryonic genetic oscillators into spatiotemporal wave patterns, *Cell* 164 (2016) 656–667.
- [102] H. Shimojo, R. Kageyama, Making waves toward the shore by synchronicity, *Dev. Cell* 36 (2016) 358–359.
- [103] G. Pernelle, W. Nicola, C. Clopath, Gap junction plasticity as a mechanism to regulate network-wide oscillations, *PLoS Comput. Biol.* 14 (2018) e1006025.
- [104] B.A. Miao, L. Meng, B. Tian, Biology-guided engineering of bioelectrical interfaces, *Nanoscale Horiz.* 7 (2022) 94–111.
- [105] F. Varela, J.P. Lachaux, E. Rodriguez, J. Martinerie, The brainweb: phase synchronization and large-scale integration, *Nat. Rev. Neurosci.* 2 (2001) 229–239.
- [106] C.N. Allen, M.N. Nitabach, C.S. Colwell, Membrane currents, gene expression, and circadian clocks, *Cold Spring Harb. Perspect. Biol.* 9 (2017) a027714.
- [107] S. Berto, G.-Z. Wang, J. Germi, B.C. Lega, G. Konopka, Human genomic signatures of brain oscillations during memory encoding, *Cereb. Cortex* 28 (2018) 1733–1748.
- [108] A.E. Pereda, Electrical synapses and their functional interactions with chemical synapses, *Nat. Rev. Neurosci.* 15 (2014) 250–263.
- [109] E.H. Shin, Y. Li, U. Kumar, H.V. Sureka, X. Zhang, C.K. Payne, Membrane potential mediates the cellular binding of nanoparticles, *Nanoscale* 5 (2013) 5879–5886.
- [110] Y. Wang, X. Han, Z. Cui, D. Shi, Bioelectricity, its fundamentals, characterization methodology, and applications in nano-bioprobing and cancer diagnosis, *Adv. Biosyst.* 3 (2019) e1900101.
- [111] Y. Zhou, C.O. Wong, K.J. Cho, D. van der Hoeven, H. Liang, D.P. Thakur, J. Luo, M. Babic, K.E. Zinsmaier, M.X. Zhu, H. Hu, K. Venkatachalam, J.F. Hancock, Membrane potential modulates plasma membrane phospholipid dynamics and K-Ras signaling, *Science* 349 (2015) 873–876.
- [112] A. Acardi, Lipids link ion channels and cancer, *Science* 349 (2015) 789–790.
- [113] M. Kerszberg, L. Wolpert, Specifying positional information in the embryo: looking beyond morphogens, *Cell* 130 (2007) 205–209.
- [114] A. Li, J.H. Cho, B. Reid, C.-C. Tseng, L. He, P. Tan, C.-Y. Yeh, P. Wu, Y. Li, R.B. Widellitz, Y. Zhou, M. Zhao, R.H. Chow, C.-M. Chuong, Calcium oscillations coordinate feather mesenchymal cell movement by SHH dependent modulation of gap junction networks, *Nature Commun.* 9 (2018) 5377.
- [115] A. Hanson, Spontaneous electrical low-frequency oscillations: a possible role in hydra and all living systems, *Philos. Trans. R. Soc. B* 376 (2021) 20190763.
- [116] D.S. Adams, J.M. Lemire, R.H. Kramer, M. Levin, Optogenetics in Developmental Biology: using light to control ion flux-dependent signals in *Xenopus* embryos, *Int. J. Dev. Biol.* 58 (2014) 851–861.
- [117] B.T. Chernet, D.S. Adams, M. Lobikin, M. Levin, Use of genetically encoded, light-gated ion translocators to control tumorigenesis, *Oncotarget* 7 (2016) 19575–19588.
- [118] J. Delile, M. Herrmann, N. Peyri  ras, R. Doursat, A cell-based computational model of early embryogenesis coupling mechanical behaviour and gene regulation, *Nature Commun.* 8 (2017) 13929.
- [119] C. Collinet, T. Lecuit, Programmed and self-organized flow of information during morphogenesis, *Nat. Rev. Mol. Cell. Biol.* 22 (2021) 245–265.
- [120] B.B. Silver, A.E. Wolf, J. Lee, M.F. Pang, C.M. Nelson, Epithelial tissue geometry directs emergence of bioelectric field and pattern of proliferation, *Mol. Biol. Cell.* 31 (2020) 1691–1702.
- [121] R. Yuste, M. Levin, New Clues about the Origins of Biological Intelligence, *Scientific American*, 2021.
- [122] R.V. Sol  , S. Valverde, Spontaneous emergence of modularity in cellular networks, *J. R. Soc. Interface* 5 (2008) 129–133.
- [123] M. Levin, Technological approach to mind everywhere: An experimentally-grounded framework for understanding diverse bodies and minds, *Front. Syst. Neurosci.* 16 (2022) 768201.
- [124] V.P. Pai, C.J. Martyniuk, K. Echeverri, S. Sundelacruz, D.L. Kaplan, M. Levin, Genome-wide analysis reveals conserved transcriptional responses downstream of resting potential change in *Xenopus* embryos, axolotl regeneration, and human mesenchymal cell differentiation, *Regeneration* 3 (2015) 3–25.
- [125] R. Yuste, From the neuron doctrine to neural networks, *Nat. Rev. Neurosci.* 16 (2015) 487–497.
- [126] C.D. McCaig, B. Song, A.M. Rajnicek, Electrical dimensions in cell science, *J. Cell Sci.* 122 (2009) 4267–4276.
- [127] J. Carvalho, A computational model of cell membrane bioelectric polarization and depolarization, connected with cell proliferation, in different tissue geometries, *J. Theoret. Biol.* 557 (2023) 111338.